Course No. Chem. 340F
Chemical Spectroscopy
Full Marks: 70+20+10=100 (1 Unit, 4 Credits)

Learning Objectives:

Learning objectives of this course are to

- realize the interaction between electromagnetic radiation and matter.
- achieve knowledge on the fundamental principles of spectroscopic techniques.
- prepare students for advanced studies involving spectroscopic analysis of organic compounds.
- Attain concepts of absorption, emission and chemical shift.
- Understand basics of vibrational, rotational and other types of transitions.

Course Content

1. Interaction of Electromagnetic Radiation with Atoms and Molecules
2. Microwave Spectroscopy
3. Vibrational Spectroscopy
4. Raman Spectroscopy
5. Electronic Spectroscopy of Atoms
6. Electronic Spectroscopy of Molecules
7. Spin Resonance Spectroscopy

Reference Books

1. Fundamentals of Molecular Spectroscopy, C.N. Banwell and E.M. Mccash, Tata McGraw-Hill
2. Introduction to Molecular Spectroscopy, G.M. Barrow
3. Modern Spectroscopy, J. Michael Hollas, Willey
4. Spectroscopy (Atomic and Molecular), G. Chatwal & Shan Anand

Interaction of Electromagnetic Radiation with Atoms and Molecules

What Is Spectroscopy?

Spectroscopy is the ***study*** of the **interaction of electromagnetic radiation with matter**.

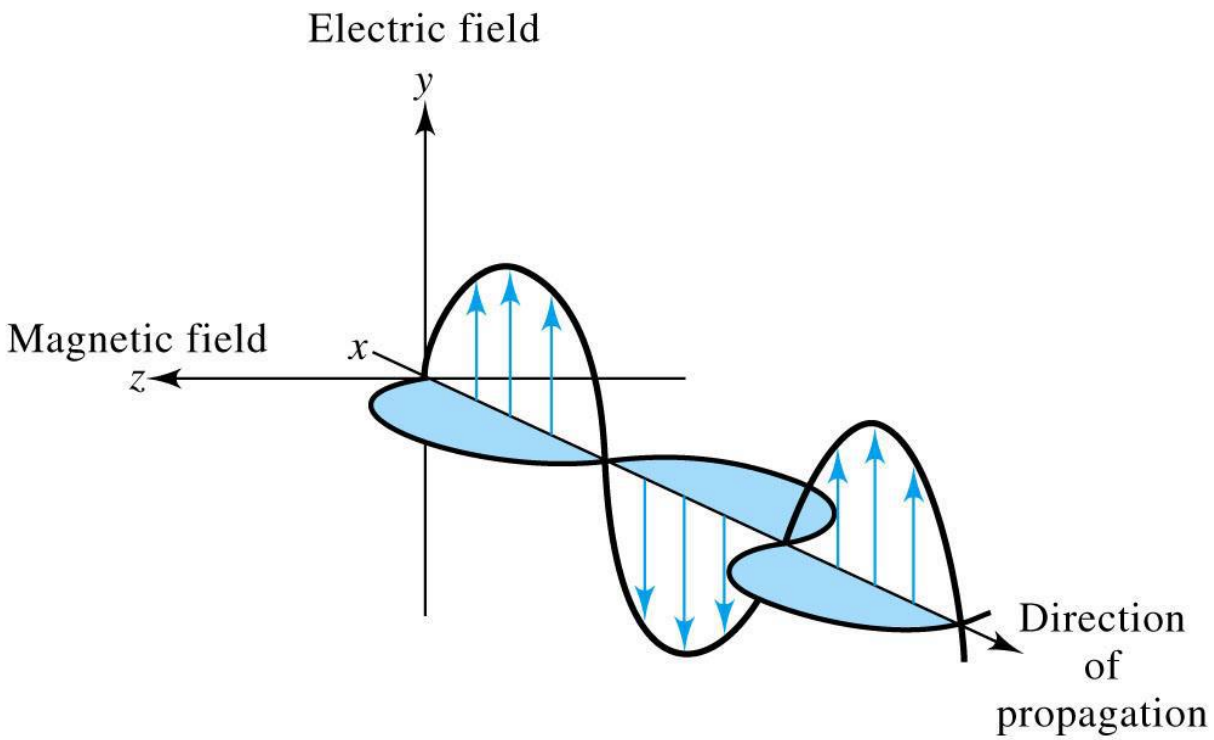
The detection and analysis of the electromagnetic radiation absorbed or emitted by species

What Is a Spectrum/spectra?

A spectrum is a plot of the intensity as a function of wavelength (or frequency or wave number, etc.) of the radiation emitted or absorbed by an atom or molecule.

Electromagnetic radiation

What Is Electromagnetic radiation (EMR)?



EMR consists of oscillating electric and magnetic fields directed perpendicular to the direction of propagation of the wave.

Electromagnetic waves are produced whenever charged particles are accelerated or whenever positive and negative charge oscillates with respect to each other.

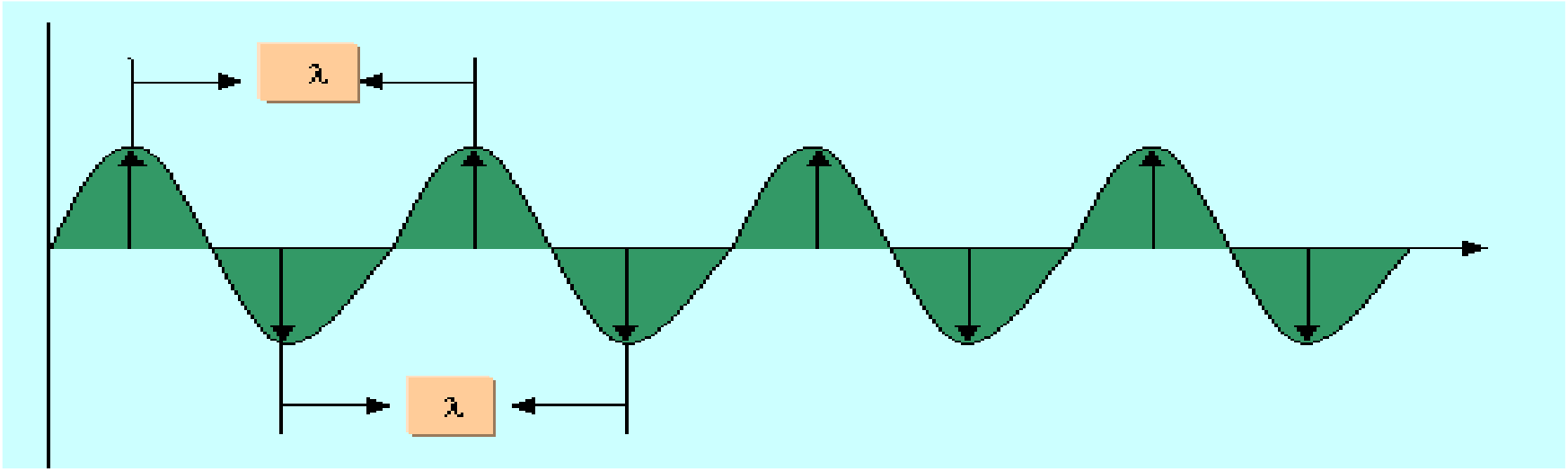
Ex: Radio waves, microwaves, IR, Visible light, UV light, X-rays and γ-rays

Electromagnetic radiation

The wave nature of an electromagnetic radiation is characterized by the following wave properties.

Wavelength (λ)

The wavelength of a wave is the distance between two consecutive maxima or two consecutive minima on the wave.



Time Period (T)

The time required for one wavelength to pass a fixed point in space.

It is expressed in seconds

Frequency (υ)

The number of wavelengths passing a fixed point in space per second.

$$\upsilon = 1/T$$

It is expressed in hertz (Hz), kilohertz (kHz), megahertz (MHz)

The frequency of the wave is directly proportional to the energy of the wave.

Wavenumber ($\bar{\upsilon}$)

It is defined as the number of wavelengths which can be accommodated in 1 cm length along the direction of propagation. Hence

$$(\bar{\upsilon}) = 1/\lambda$$

It is generally expressed in unit of cm^{-1}.

This is an important property especially in the study of infrared spectroscopy.

Wave-Particle duality of EMR

The classical wave model

- The classical wave model describes electromagnetic radiation as waves that has a wavelength, frequency, velocity, and amplitude.

- These properties of electromagnetic radiation can explain classical characteristics of electromagnetic radiation like reflection, refraction, diffraction, interference, etc.

- However, the wave model can not explain the phenomena of absorption and emission of radiation by atomic or molecular species and necessitated the suggestion that radiation has a particle nature.

Particle Model and Quantum Theory

- To explain these phenomena, Max Plank (in 1900) developed a theory called Planck's quantum theory of radiation.
- According to this theory, EM radiation made up of discrete packets of energy, called quanta, which are shot out with the velocity of light.
- Thus energy changes take place only discontinuously and discretely, always as an integral multiple of small unit of energy, hυ (i.e. photon).
- In 1905, Albert Einstein supplemented Planck's theory, EM waves can behave as a particle called a photon whose energy depends on the frequency of the waves.

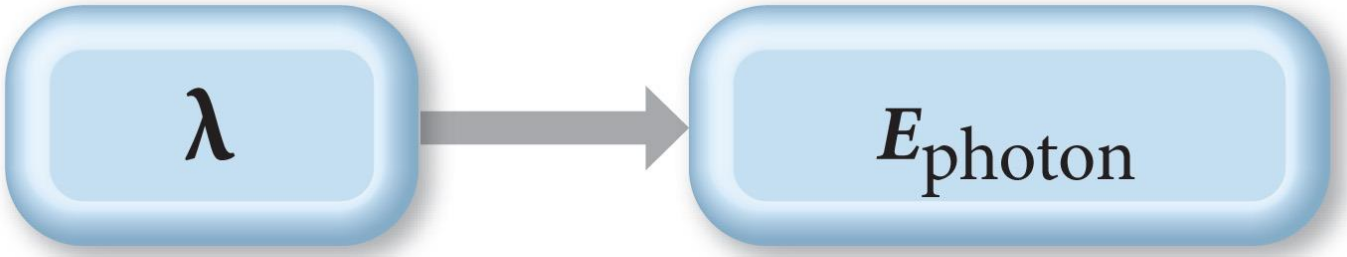
$$\mathbf{E = h\upsilon = hc/\lambda = hc\bar{\upsilon}}$$

h is Planck's constant, which has a value of 6.626×10^{-34} Js.

Quantization of Energy

- Max Planck (1900) developed a model that explained light as a quantization of energy. Energy of a light wave is present in bundles of energy called photons; the energy is said to be quantized into the photons.

- Therefore, light must be regarded as having a dual nature; in some cases light acts as a wave; in others it acts like a particle.



$$E = \frac{hc}{\lambda}$$

Energy is Quantized by *h*



Planck, 1900

Planck suggests that radiation (light, energy) can only come in quantized packets that are of size *hν.*

Energy (J) → $E = h\upsilon$

Planck's constant
h = 6.626 × 10^{-34} J·s

Frequency (s^{-1})

Note that we can specify the energy by specifying *any one* of the following:

1. The frequency, υ (units: Hz or s^{-1}): $E = h\upsilon$

2. The wavelength, λ, (units: m or cm or mm): $E = \frac{hc}{\lambda}$

   Recall: $\upsilon\lambda = c$

3. The wavenumber, $\tilde{\upsilon}$ (units: cm^{-1} or m^{-1}) $E = hc\tilde{\upsilon}$

   Recall: $\tilde{\upsilon} = \frac{1}{\lambda}$

Example

Red light has wavelength (λ) = 700 nm. What would be its energy and frequency (*υ*)?

Solution:

$\lambda = 700 \text{ nm} \times (1 \text{ m}/1 \times 10^{9} \text{ nm}) = 7.00 \times 10^{-7} \text{ m}$

$c = 3.00 \times 10^{8} \text{ ms}^{-1}$; $h = 6.626 \times 10^{-34} \text{ Js}$

We know that $\upsilon = c/\lambda$.

$\upsilon = (3.00 \times 10^{8} \text{ ms}^{-1})/(7.00 \times 10^{-7} \text{ m})$

$\upsilon = 4.29 \times 10^{14} \text{ s}^{-1}$

$E = h\upsilon$

$= (6.626 \times 10^{-34} \text{ Js}) \times (4.29 \times 10^{14} \text{ s}^{-1})$

$= 2.84 \times 10^{-19} \text{ J}$

The laser light used to read a CD has a wavelength of 785 nm. Determine the energy and frequency at 785 nm.

Solution:

$$\lambda = 785 \text{ nm} \times (1 \text{ m}/1 \times 10^{9} \text{ nm}) = 7.85 \times 10^{-7} \text{ m}$$

$$c = 3.00 \times 10^{8} \text{ ms}^{-1};\ h = 6.626 \times 10^{-34} \text{ Js}$$

We know that

$$\upsilon = c/\lambda$$

$$\upsilon = (3.0 \times 10^{8} \text{ ms}^{-1})/(7.85 \times 10^{-7} \text{ m}) = 3.82 \times 10^{14} \text{ s}^{-1}$$

$$E = h\upsilon$$

$$= (6.63 \times 10^{-34} \text{ J s}) \times (3.82 \times 10^{14} \text{ s}^{-1})$$

$$= 2.53 \times 10^{-19} \text{ J}$$

➢**So:** **Light with large λ (small *υ*) has a small energy and light with a short λ (large *υ*) has a large energy.**

Calculate the energy associated with 1.00 mol of photons of green light (555 nm).

Solution:

$\lambda = 555 \text{ nm} \times (1 \text{ m}/1 \times 10^{9} \text{ nm}) = 5.55 \times 10^{-7} \text{ m}$; $c = 3.00 \times 10^{8} \text{ ms}^{-1}$; $h = 6.626 \times 10^{-34} \text{ Js}$

1. Determine frequency (υ) $\quad \upsilon = c/\lambda$

$= (3.00 \times 10^{8} \text{ ms}^{-1})/(5.55 \times 10^{-7} \text{ m})$

$= 5.40 \times 10^{14} \text{ s}^{-1}$

2. Calculate energy

$E = h\upsilon$

$= (6.63 \times 10^{-34} \text{ Js})(5.40 \times 10^{14} \text{ s}^{-1})$

$= 3.58 \times 10^{-19} \text{ J per photon}$

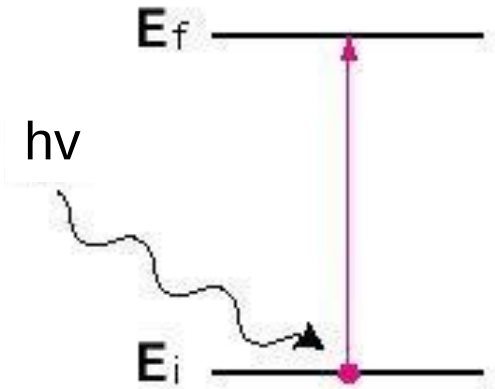
3. Determine the energy per mol.

$(3.58 \times 10^{-19} \text{ J/photon})(6.02 \times 10^{23} \text{ photons/mol})$

$= 2.16 \times 10^{5} \text{ Jmol}^{-1}$ or 216 kJ mol^{-1}

This is in the range of energies that can break bonds.

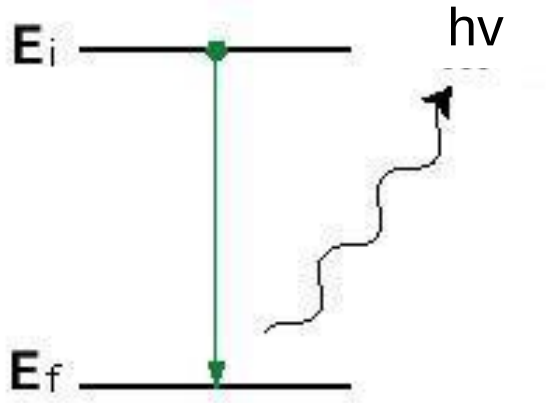
Absorption/Emission of Photons and Conservation of Energy



Absorption

When atom absorbs energy of photon to promote electron to higher energy orbital.

$$E_f - E_i = hv$$



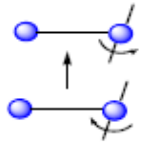
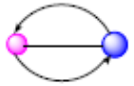
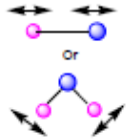
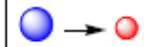
Emission

When atom emits energy as photon as electron falls from higher energy orbital to lower energy orbital.

$$E_i - E_f = hv$$

Regions of Spectrum

EMR is divided into different regions based on the type of atomic or molecular transition that gives rise to the absorption or emission of photons.

	Change of spin		Change of orientation	Change of configuration	Change of electron distribution		Change of nuclear configuration
	radiowave		Microwave	Infra-red	Visible and UV	X-ray	γ- ray
	NMR	ESR					
							
$\bar{\nu}$ in cm^{-1}	10^{-2}		1	100	10^4	10^6	10^8
λ	10m	100cm	1cm	100μm	1μm	10nm	100pm
ν in Hz	3×10^6	3×10^8	3×10^{10}	3×10^{12}	3×10^{14}	3×10^{16}	3×10^{18}
E in J/mol	10^{-3}	10^{-1}	10	10^3	10^5	10^7	10^9

Radiofrequency region: $3 \times 10^6 - 3 \times 10^{10}$ Hz; 10 m – 1 cm wavelength. Nuclear magnetic resonance (nmr) and electron spin resonance (esr) spectroscopy. The energy change involved is that arising from the reversal of spin of a nucleus or electron, and is of the order 0.001-10 j/mol.

Microwave region: $3 \times 10^{10} - 3 \times 10^{12}$ Hz; 1 cm – 100 μm wavelength. Rotational spectroscopy. Separations between the rotational levels of molecules are of the order of hundreds of joules per mole.

Infra-red region: $3 \times 10^{12} - 3 \times 10^{14}$ Hz; 100 μm – 1 μm wavelength. Vibrational spectroscopy. One of the most valuable spectroscopic regions for the chemist. Separations between levels are some 10^4 joules per mole.

Visible and ultra-violet regions: $3 \times 10^{14} - 3 \times 10^{16}$ Hz; 1 μm – 10 nm wavelength. Electronic spectroscopy. The separations between energies of valence electrons are some hundreds of kilojoules per mole.

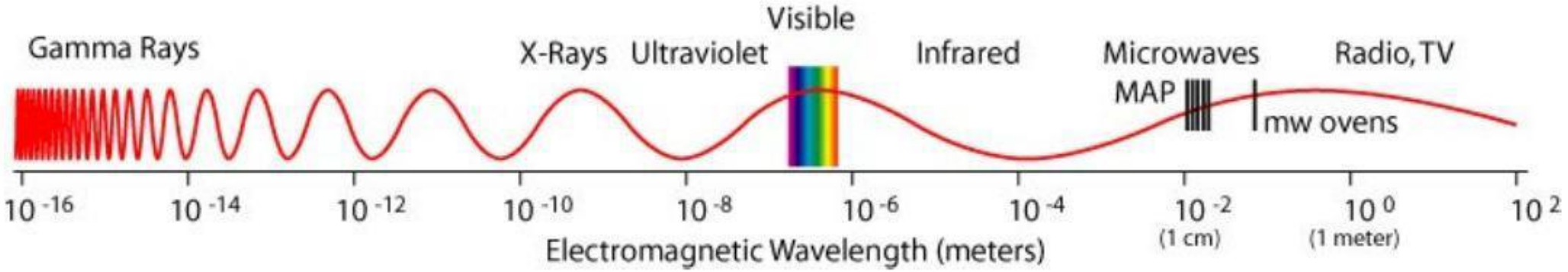
X-ray region: $3 \times 10^{16} - 3 \times 10^{18}$ Hz; 10 nm – 100 pm wavelength. Energy changes involving the inner electrons of an atom or a molecule, which may be of order ten thousand kilojoules.

γ-ray region: $3 \times 10^{18} - 3 \times 10^{20}$ Hz; 100 pm – 1 pm wavelength. Energy changes involve the rearrangement of nuclear particles, having energies of $10^9 - 10^{11}$ joules per gram atom.

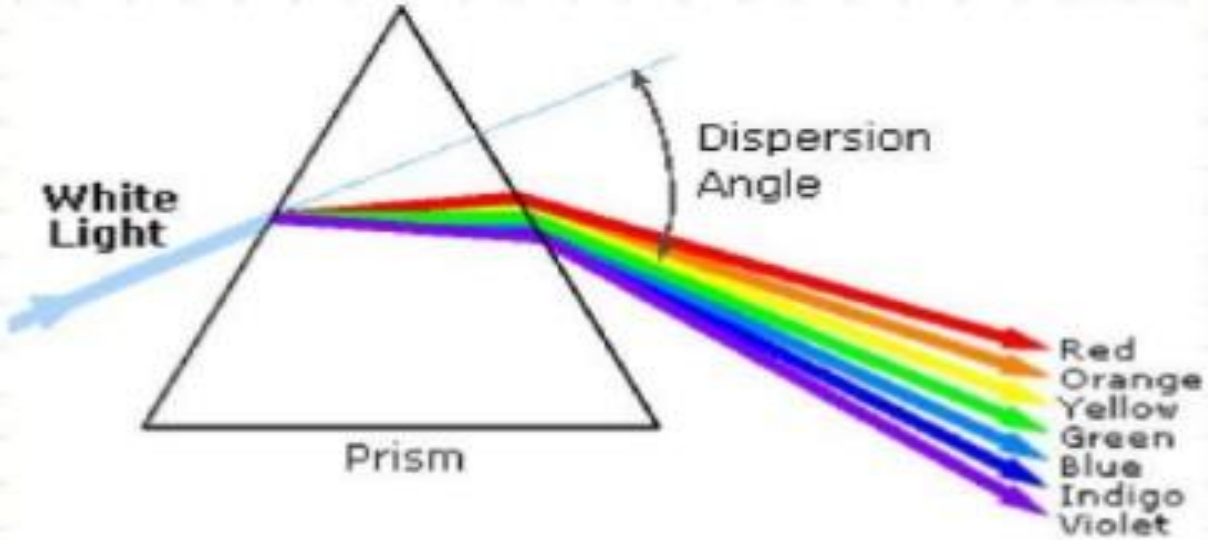
Electromagnetic Spectrum

Arrangement of EM radiations in order of their wavelengths or frequencies

The Sun produces full spectrum of electromagnetic radiations.



Electromagnetic Radiation



Violet	400 - 420 nm	Yellow	570 - 585 nm
Indigo	420 - 440 nm	Orange	585 - 620 nm
Blue	440 - 490 nm	Red	620 - 780 nm
Green	490 - 570 nm		

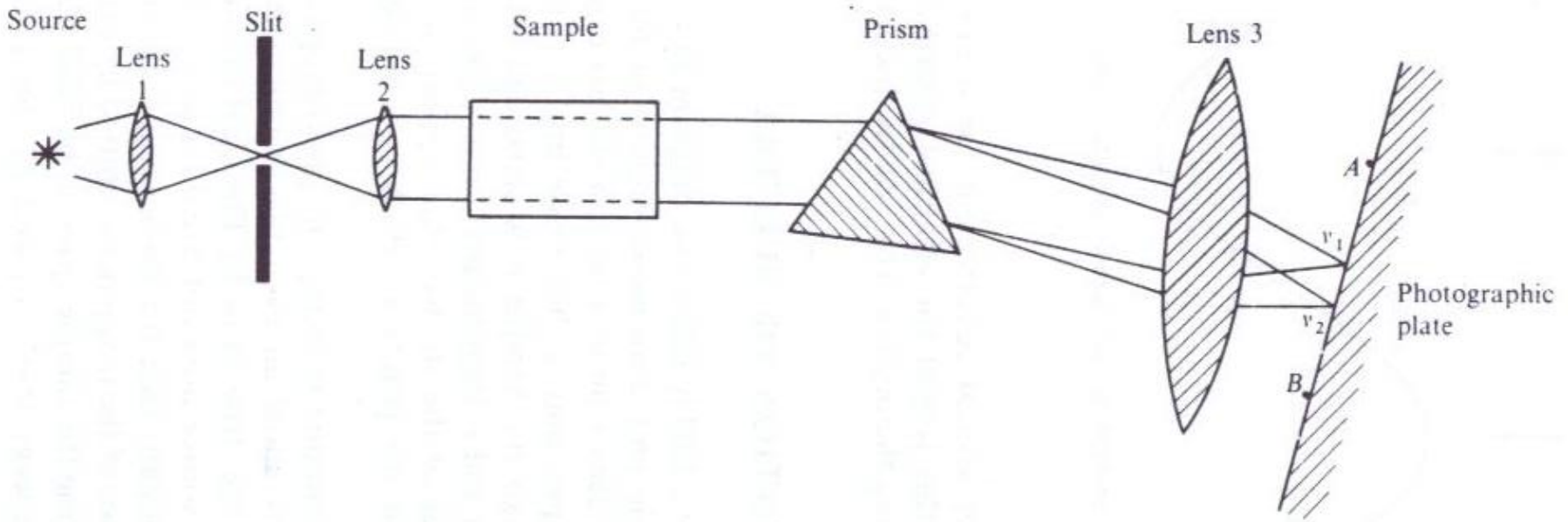
Principles of Spectroscopy

The principle is based on the measurement of spectrum of a sample containing atoms / molecules.
Spectrum is graph of intensity of absorbed or emitted radiation by sample versus frequency (υ) or wavelength (λ).
Spectrometer is an instrument design to measure the spectrum of a compound.

Absorption Spectroscopy:
An analytical technique which concerns with the measurement of absorption of electromagnetic radiation. e.g. UV (185 – 400 nm) / Visible (400 – 800 nm) Spectroscopy, IR Spectroscopy (0.76 – 15 µm)

Emission Spectroscopy:
An analytical technique in which emission (of a particle or radiation) is dispersed according to the some properties of emission and the amount of dispersion is measured.

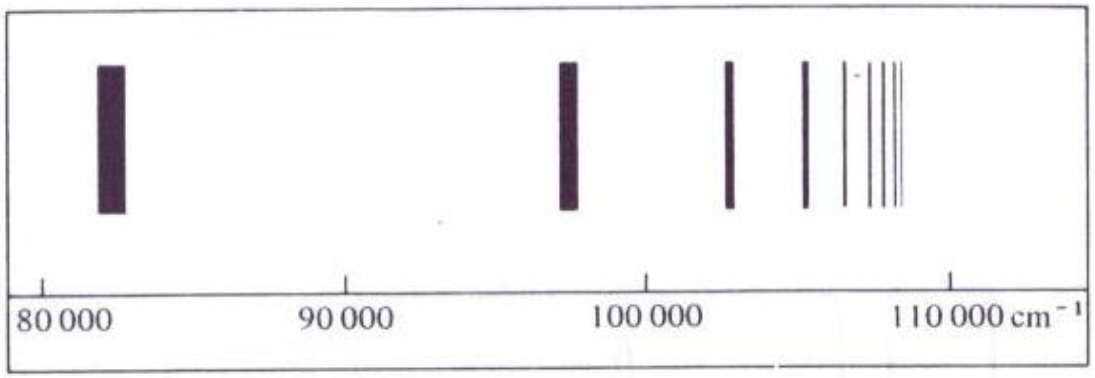
Schematic diagram of a spectrometer suitable for operation in the visible region



A white source is focused by lens 1 on to a narrow slit (arranged perpendicularly to the plane of the paper) and is then made into a parallel beam by lens 2. After passing through the sample it is separated into its constituent frequencies by a prism and is then focused on to a photographic plate by lens 3; the vertical image of the slit will thus appear on the plate. Rays have been drawn to show the points at which two frequencies, v_1 and v_2, are focused.

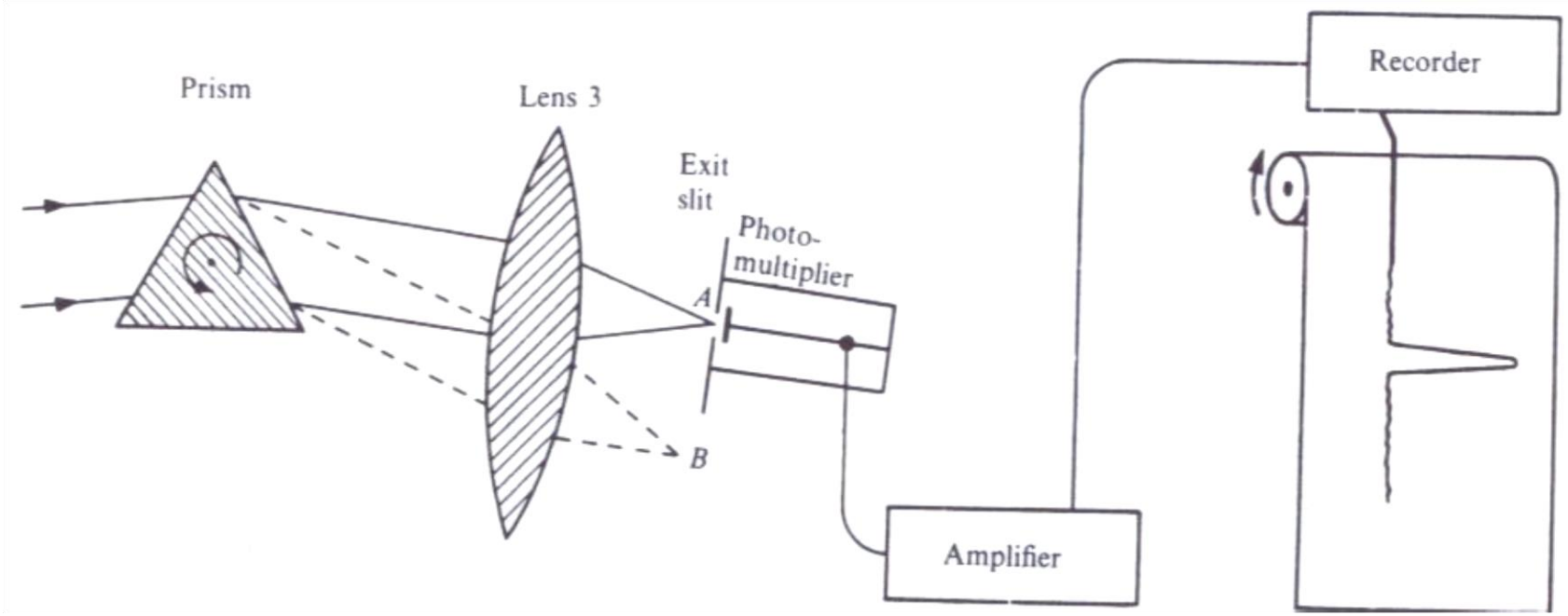
If the sample space filled with a substance having only two possible energy levels E_1 and E_2, the photographic plate will show a blackening at all points except at the frequency $v = (E_1 - E_2)/h$, since energy at this frequency will have been absorbed by the sample in raising each molecule from state 1 to state 2. Further if there are many possible energy levels, E_1, E_2,.....E_j, E_k….. available to the sample, a series of absorption lines will appear on the photographic plate at frequency given by $v = (E_j - E_k)/h$. A typical spectrum may then appear as

Schematic diagram of the absorption spectrum of atomic hydrogen recorded on a photographic plate

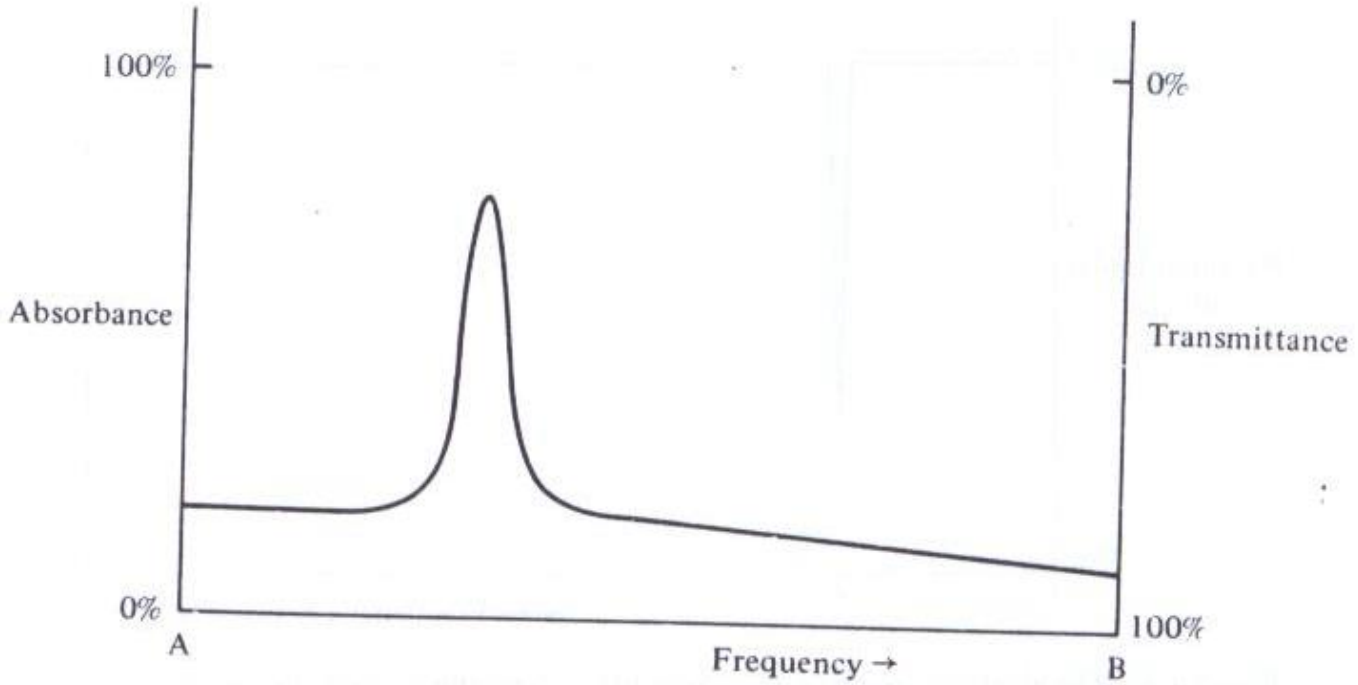


At this point it may be helpful to consider what happens to the energy absorbed in the short of process described above. In the ultra-violet, visible, and infra-red regions it is an experimental fact that a given sample continues to show an absorption spectrum for as long as we care to irradiate it- in other words, a finite number of sample molecules appear to be capable of absorbing an infinite amount of energy. Simply the molecules must be able to rid themselves of the absorbed energy.

In modern spectrometers the detector is rarely the simple photographic plate of the previous Figure. One of the most sensitive and useful devices in the visible and ultra-violet region is the photomultiplier tube, consisting of a light-sensitive surface which emits electrons when light falls upon it. The tiny electron current may be amplified and applied to an ammeter or pen recorder. The spectrometer would then appear somewhat as in next Figure

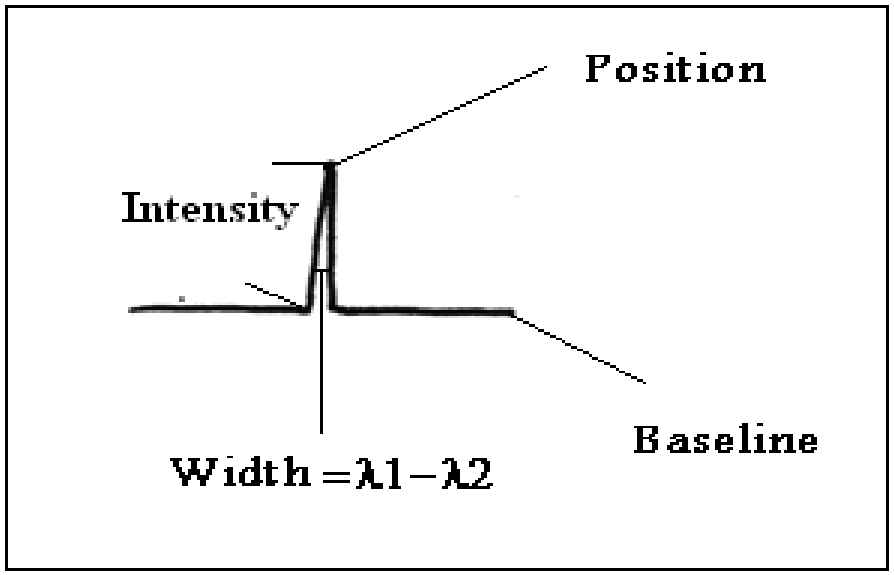


Schematic diagram of a spectrometer employing a photomultiplier or other sensitive element as detector and recording the spectrum graphically on chart paper

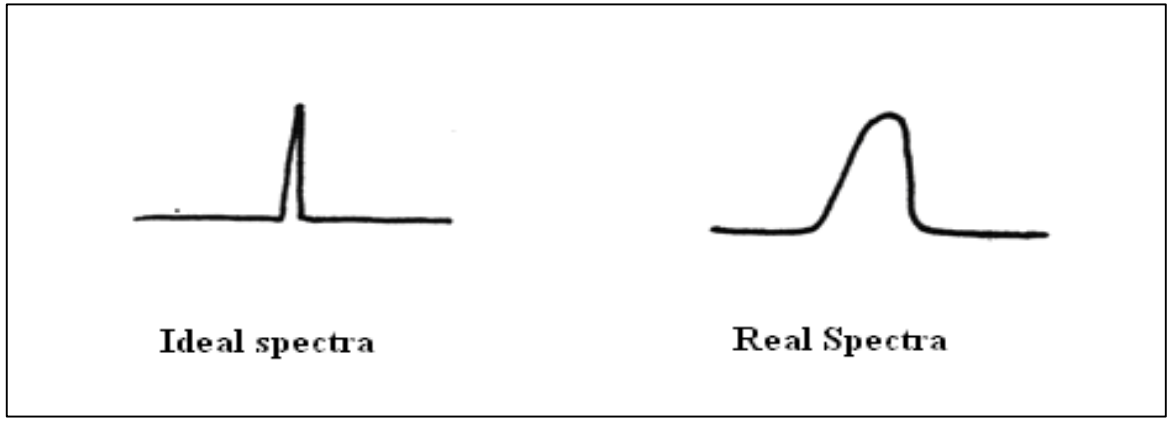


The usual appearance of the spectrum of a molecule undergoing a single transition; here the background is no longer constant and the absorption region is of finite width.

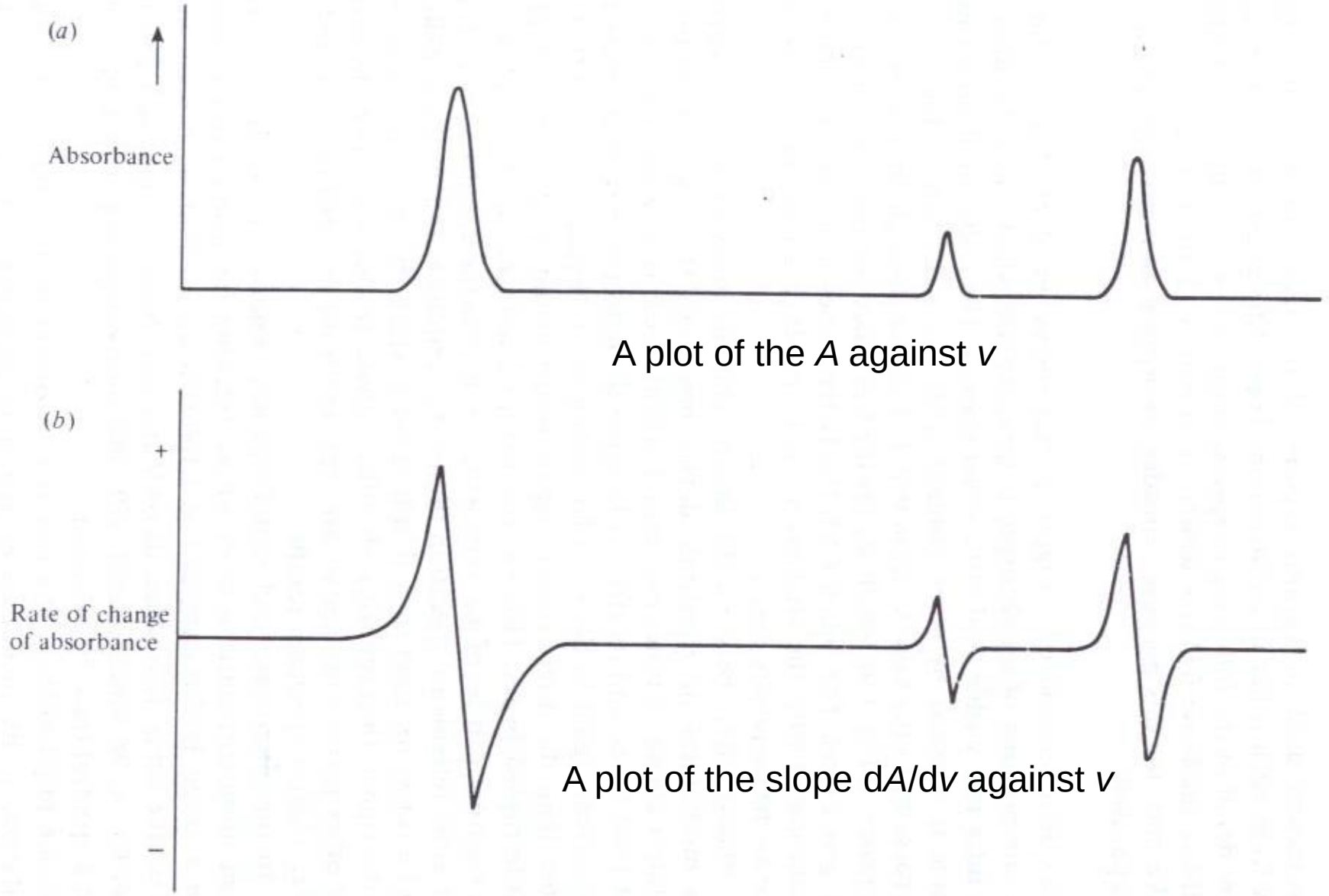
Representation of Spectra



Generally the spectral band is characterized by position, width, and intensity. The position of the spectral bands, or the frequencies at which the molecule absorb, depends on its structural features (functional group) as well as on its environments.



The two most important features of spectral lines are -
their height (~ intensity) and their width (Δn)



To illustrate the relation between absorption and derivative spectra: in (a) the absorption nuclear magnetic resonance spectrum of benzyl alcohol, $C_6H_5CH_2OH$ is shown, and in (b) the derivative (or dispersion) spectrum of the same molecule.

Signal-to-Noise: Resolving Power

Signal-to-Noise Ratio

Since almost all modern spectrometers use some form of electronic amplification to magnify the signal produced by the detector, every recorded spectrum has a background of random fluctuations caused by spurious electronic signals produced by the detector, or generated in the amplifying equipment. These fluctuations are usually referred to as 'noise'. In order that a real spectral peak should show itself as such and be sufficiently distinguished from the noise, it must have an intensity some three or four times that of the noise fluctuations (a signal-to-noise ratio of three or four). This requirement places a lower limit on the intensity of observable signals. In Sec. 1.9 we refer briefly to a computer-averaging technique by which it is possible to improve the effective signal-to-noise ratio.

Resolving Power

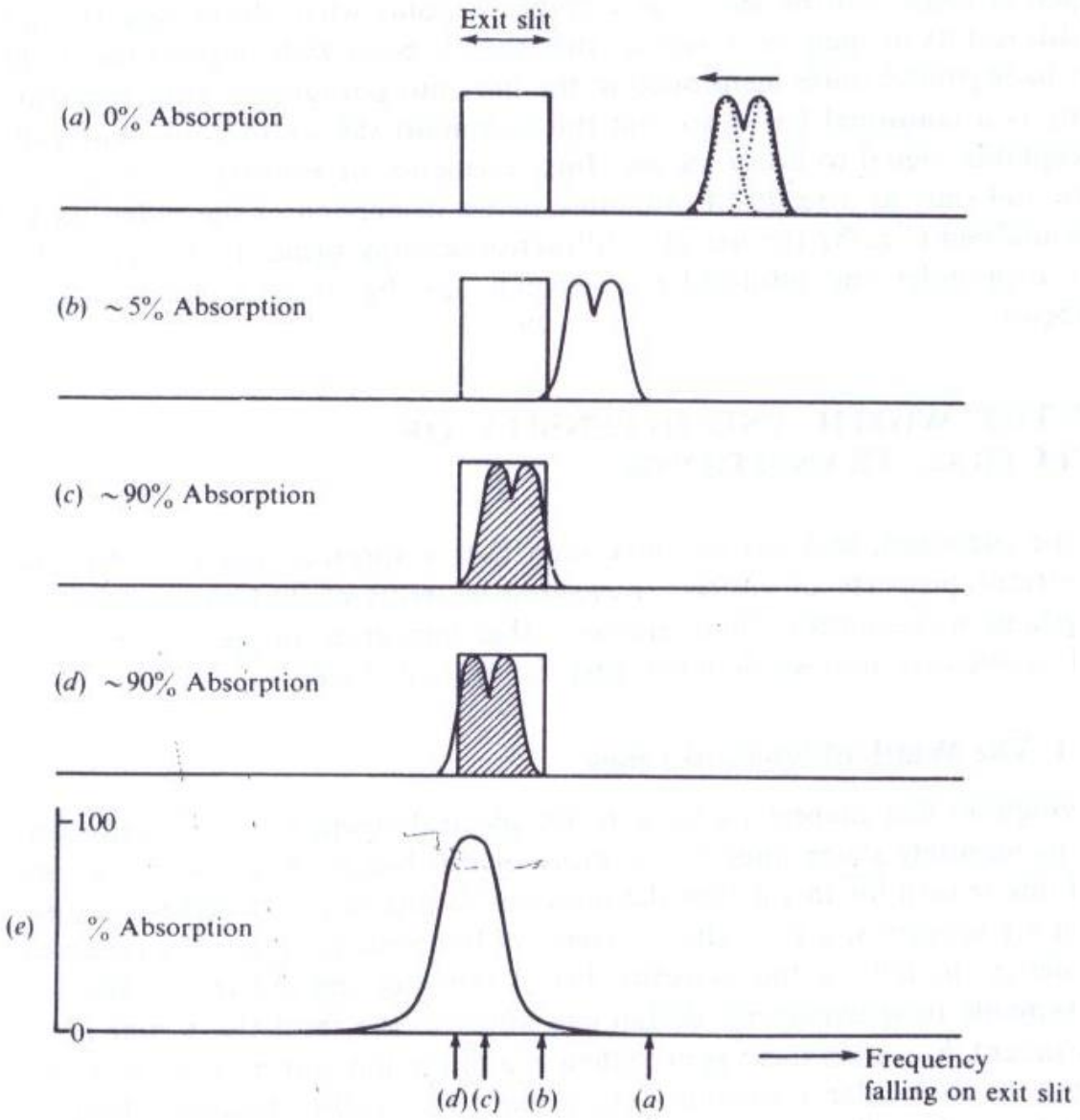


Figure 1.17 Illustrating the relation between slit width and resolving power: see text for discussion.

This is a somewhat imprecise concept which can, however, be defined rather arbitrarily and is often used as a measure of the performance of a spectrometer. We shall here consider it in general terms only.

No molecular absorption takes place at a single frequency only, but always over a spread of frequencies, usually very narrow but sometimes quite large (see Sec. 1.7); it is for this reason that we have up to now drawn spectra with broadened line shapes (cf. Fig. 1.14(*a*)).

Let us consider two such lines close together, as on the right of Fig. 1.17(*a*): the dotted curve represents the absorption due to each line separately, the full line their combined absorption. We shall first take the exit slit width to be larger than the separation between the lines. Scanning the spectrum plainly involves moving the twin absorbance peaks steadily to the left so that they pass across the exit slit and into the detector; the situation at successive stages is shown in (*b*), (*c*), and (*d*) of Fig. 1.17, the shaded area showing the amount of absorbance which the detector would register. At (*e*) of this figure, the absorbance is plotted against frequency, together with the approximate positions of stages (*a*), (*b*), (*c*), and (*d*).

It is quite evident that the separation between the lines has disappeared under these conditions—the lines are not *resolved*. It is equally evident that the use of a much narrower slit would result in their resolution—the *resolving power* would be increased. In fact, provided the slit width is less than the separation between the lines, the detector output will show a minimum between them.

However, it must be remembered that a narrower slit allows less total energy from the beam to reach the detector and consequently the intrinsic signal strength will be less. There comes a point when decreasing the slit width results in such weak signals that they become indistinguishable from the background noise mentioned in the previous paragraph. Thus spectroscopy is a continual battle to find the minimum slit width consistent with acceptable signal-to-noise values. Improvements in resolving power may arise not only as a result of obtaining better dispersion of the radiation by the analyser (e.g., by the use of a diffraction grating rather than a prism for the ultra-violet and infra-red regions) but also by using a more sensitive detector.

The Width and Intensity of Spectral Transitions

A spectral transition has the important property of position, measured in terms of its frequency, wavelength or wavenumber; there are two other important properties, its width and its intensity.

The Width of Spectral Lines

Several factors contribute to this.

1. *Collision broadening.* Atoms or molecules in liquid and gaseous phases are in continual motion and collide frequently with each other. These collisions inevitably cause some deformation of the particles and hence perturb, to some extent, the energies of at least the outer electrons in each. This immediately gives a possible explanation for the width of visible and ultra-violet spectral lines, since these deal largely with transitions between outer electronic shells. Equally vibrational and rotational spectra are broadened since collisions interfere with these motions too. In general, molecular interactions are more severe in liquids than in gases, and gas-phase spectra usually exhibit sharper lines than those of the corresponding liquid.

   In the case of solids, the motions of the particles are more limited in extent and less random in direction, so that solid-phase spectra are often sharp but show evidence of interactions by the splitting of lines into two or more components.

2. *Doppler broadening.* Again in liquids and gases the motion of the particles causes their absorption and emission frequencies to show a Doppler shift; since the motion is random in a given sample, shifts to both high and low frequencies occur and hence the spectral line is broadened. In general, for liquids collision broadening is the most important factor, whereas for gases, where collision broadening is less pronounced, the Doppler effect often determines the natural line width.

3. *Heisenberg uncertainty principle.* Even in an isolated, stationary molecule or atom the energy levels are not infinitely sharp, due to the operation of a fundamental and very important principle, the Uncertainty Principle of Heisenberg. In effect this says that, if a system exists in an energy state for a limited time δt seconds, then the energy of that state will be uncertain (fuzzy) to an extent δE where

$$\delta E \times \delta t \approx h/2\pi \approx 10^{-34} \text{ J s} \tag{1.10}$$

where h is again Planck's constant. Thus we see that the lowest energy state of a system *is* sharply defined since, left to itself, the system will remain in that state for an infinite time; thus $\delta t = \infty$, and $\delta E = 0$. But, for example, the lifetime of an excited electronic state is usually only about 10^{-8} s, which gives a value for δE of about $10^{-34}/10^{-8} = 10^{-26}$ J. A transition between this state and the ground state will thus have an energy uncertainty of δE, and a corresponding uncertainty in the associated radiation frequency of $\delta E/h$, which we can write as:

$$\delta \nu = \frac{\delta E}{h} \approx \frac{h}{2\pi h \delta t} \approx \frac{1}{2\pi \delta t} \tag{1.11}$$

Thus for our example of an excited electronic state lifetime of 10^{-8} s, $\delta \nu \approx 10^8$ Hz. This apparently large uncertainty is, in fact, small compared with the usual radiation frequency of such transitions, 10^{14}–10^{16} Hz, and so the natural line width is said to be small; in fact, the apparent widths of electronic transitions are far more dependent on collision and Doppler broadening than on energy uncertainties.

The Intensity of Spectral Lines

When discussing spectral intensities there are three main factors to be considered: the likelihood of a system in one state changing to another state—the *transition probability*; the number of atoms or molecules initially in the state from which the transition occurs—the *population*; and the amount of material present giving rise to the spectrum—the *concentration* or *path length* of the sample.

1. *Transition probability.* The detailed calculation of absolute transition probabilities is basically a straightforward matter, but as it involves a knowledge of the precise quantum mechanical wave functions of the two states between which the transition occurs, it can seldom be done with accuracy and is, in any case, beyond the scope of this book. We shall generally content ourselves with qualitative statements about relative transition probabilities without attempting any detailed calculations.

At a much lower level of sophistication, however, it is often possible to decide whether a particular transition is forbidden or allowed (i.e., whether the transition probability is zero or non-zero). This process is essentially the deduction of *selection rules*, which allow us to decide between which levels transitions will give rise to spectral lines, and it can often be carried out through pictorial arguments very like those we have already used in discussing the activity or otherwise of processes in Sec. 1.3.

2. *Population of states.* If we have two levels from which transitions to a third are equally probable, then obviously the most intense spectral line will arise from the level which initially has the greater population. There is a simple statistical rule governing the population of a set of energy levels.

For example, if we have a total of N molecules distributed between two different energy states, a lower and an upper with energies E_{lower} and E_{upper}, respectively, we would intuitively expect most of the molecules to occupy the lower state. Proper statistical analysis bears this out and shows that, *at equilibrium*

$$\frac{N_{\text{upper}}}{N_{\text{lower}}} = \exp(-\Delta E/kT) \tag{1.12}$$

where $\Delta E = E_{\text{upper}} - E_{\text{lower}}$, T is the temperature in K, and k is a universal constant. The expression is known as the Boltzmann distribution, after its originator, and k, which has a value of $1{\cdot}38 \times 10^{-23}$ J K^{-1}, as Boltzmann's constant. Examples showing the use of this very important expression will recur throughout the remaining chapters.

3. *Path length of sample.* Clearly if a sample is absorbing energy from a beam of radiation, the more sample the beam traverses the more energy will be absorbed from it. We might expect that twice as much sample would give twice the absorption, but a very simple argument shows that this is not so. Consider two identical samples of the same material, S_1 and S_2, and assume that S_1 or S_2 alone absorb 50 per cent of the energy falling on them, allowing the remaining 50 per cent to pass through. If we pass a beam of initial intensity I_0 through S_1, 50 per cent of I_0 will be absorbed and the intensity of the beam leaving S_1 will be $\frac{1}{2}I_0$; if we then pass this beam through S_2 a further 50 per cent will be absorbed, and $\frac{1}{2} \times \frac{1}{2}I_0 = \frac{1}{4}I_0$ will leave S_2. Thus two 50 per cent absorptions in succession do not add up to 100 per cent but only to 75 per cent absorption. An exactly similar relationship exists between the *concentration* of a sample and the amount of energy absorption—a doubling of the concentration produces something less than a doubling of the absorption.

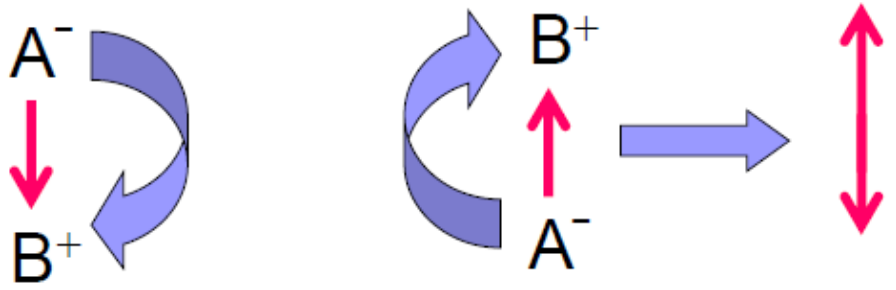
These relationships are best expressed in terms of the Beer–Lambert law, which is:

$$\frac{I}{I_0} = \exp(-\varepsilon c l) \tag{1.13}$$

where I_0 is the intensity of radiation falling on the sample, and I that part transmitted, c and l are the sample concentration and length, and ε is the extinction coefficient or absorption coefficient, which is a constant for a given type of transition (e.g., electronic, vibrational, etc.) occurring within a particular sample. Clearly ε is closely connected with the transition probability discussed above, a large probability being associated with a large ε, and vice versa.

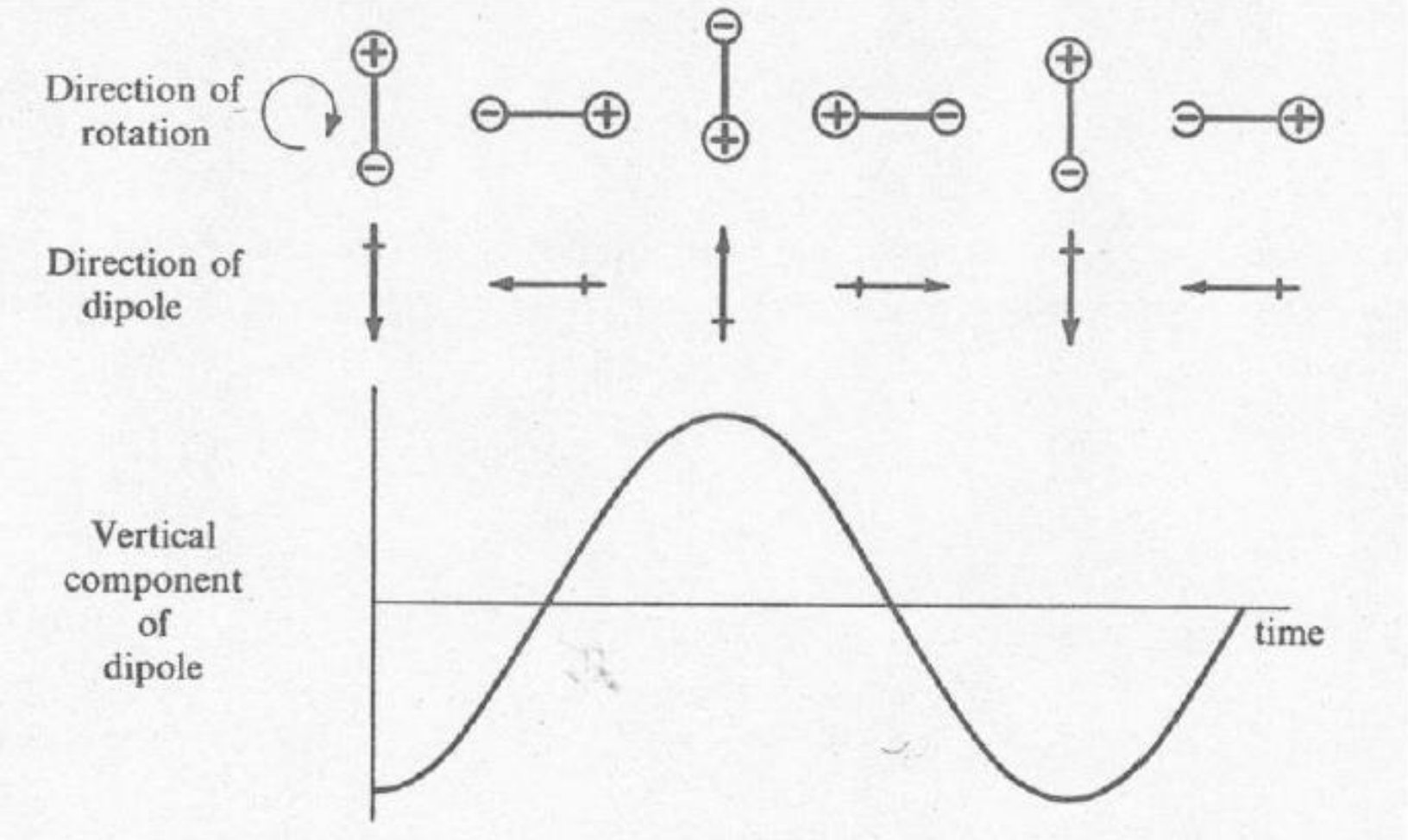
Microwave Spectroscopy

It is concerned with transitions between rotational energy levels in the molecules, the molecule gives a rotational spectrum only If it has a permanent dipole moment:



- H-Cl, and C=O give rotational spectrum (microwave active).
- H-H and Cl-Cl don't give rotational spectrum (microwave inactive).
- Rotational spectroscopy is only really practical in the gas phase where the rotational motion is quantized. In solids or liquids the rotational motion is usually quenched due to collisions between their molecules.

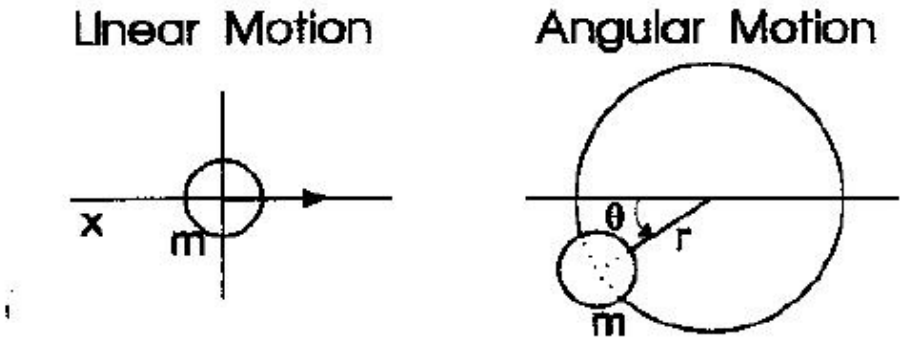
Rotational energy of a diatomic molecule



The rotation of a diatomic molecule, HCl, *showing the fluctuation in the dipole moment measured in a particular direction.*

General features of rotating system:

Rotational motion in classical mechanics
Rigid Rotors: molecules in which bonds don't distort under the stress of rotation.



Linear momentum (P) = m.v

Angular momentum (J) = $I.\omega$

Then; $I = m.r^2$

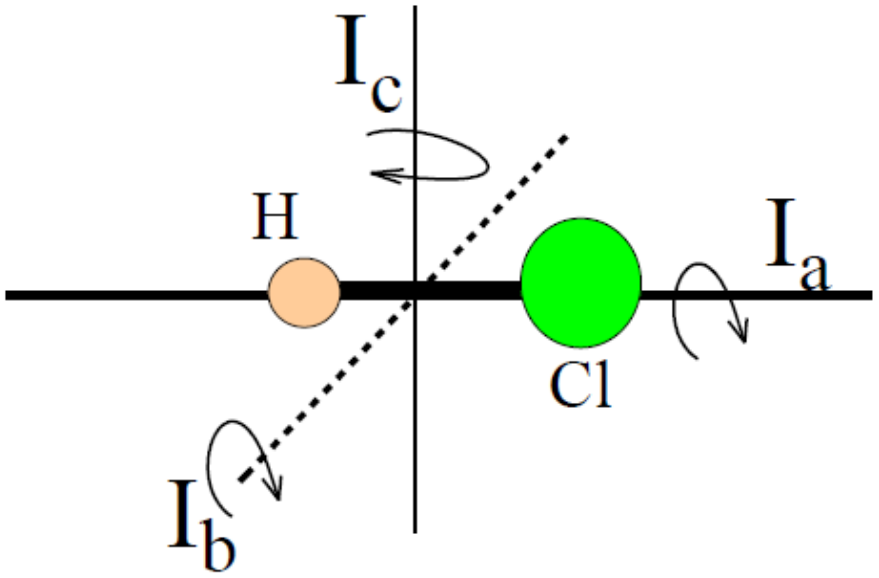
Where; the moment of inertia for a molecule (I) = $\Sigma_i m_i.r_i^2$

$$\text{Linear velocity (v)} = \frac{\text{Distance}}{\text{Time}} \qquad \text{Angular velocity } (\omega) = \frac{\text{Radians}}{\text{Time}}$$

Where: r_i is the perpendicular distance of the atom i from the axis of rotation (bond length).

Moment of inertia (I), also called mass moment of inertia which is a measure of an object's resistance to changes in its rotation rate. It is the rotational analog of mass.

A molecule can have three different moments of inertia I_a, I_b and I_c, according to the axis of rotation.

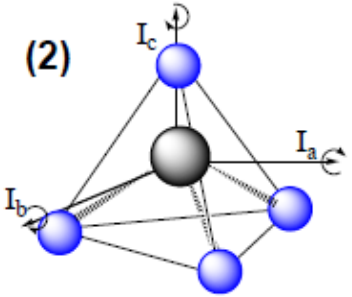


In HCl, $I_a = 0$, $I_b = I_c$.

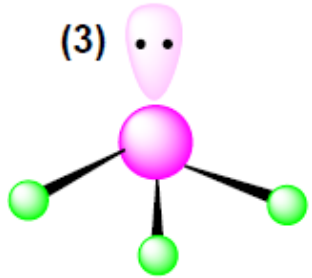
Rigid rotors are classified into four groups:

1- Linear rotors: such as diatomic or linear molecules, as H-Cl, O=C=S, acetylene and O=C=O, have; I_a=0 and I_b= I_c .

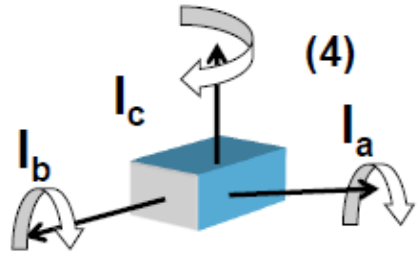
2- Spherical tops rotors: e.g. CH_4 , SiH_4 and SF_6 have three equals moment of inertia. I_a= I_b= I_c.



3- Symmetric tops rotors: e.g. NH_3 , CH_3CN and CH_3Cl, have two equal moments of inertia. , I_a= I_b ≠ I_c .



4- Asymmetric tops rotors: e.g. H_2O, CH_3OH, vinyl chloride CH_2=CHCl and formaldehyde, have three different moments of inertia. I_a ≠ I_b ≠ I_c .

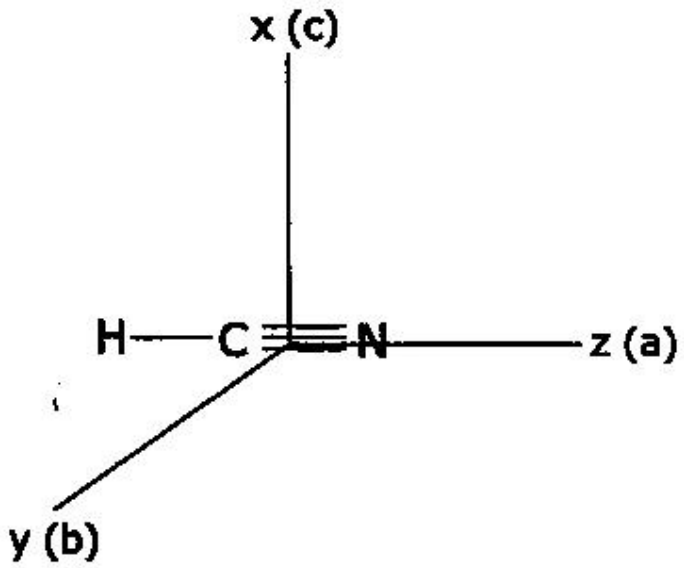


Classes of Rotating Molecules

- Molecules can be classified into five main groups depending on their moments of inertia.

1.	$I_C = I_B$, $I_A = 0$	Linear molecules
2.	$I_C = I_B = I_A$	Spherical top
3.	$I_C = I_B > I_A$	Prolate symmetric top
4.	$I_C = I_B < I_A$	Oblate symmetric top
5.	$I_C \neq I_B \neq I_A$	Asymmetric top

Linear Molecules



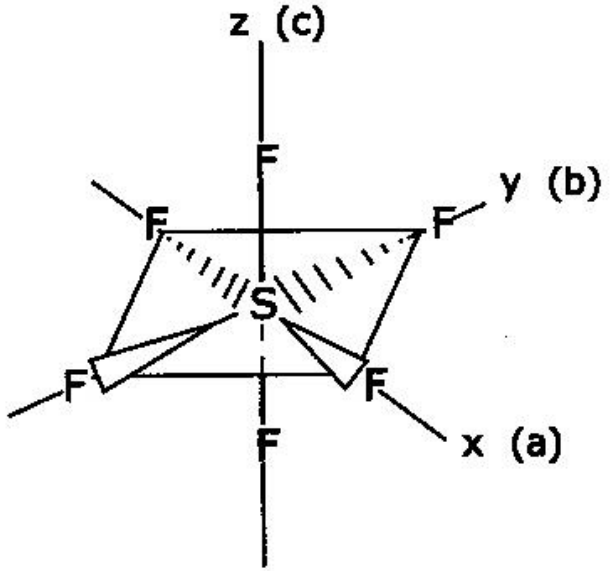
$I_C = I_B$, $I_A = 0$

Linear molecules

Other examples:

- HCl
- CO_2
- H–C≡C–H
- H–C≡C–C≡C–C≡C–H
- LiF

Spherical Top Molecules



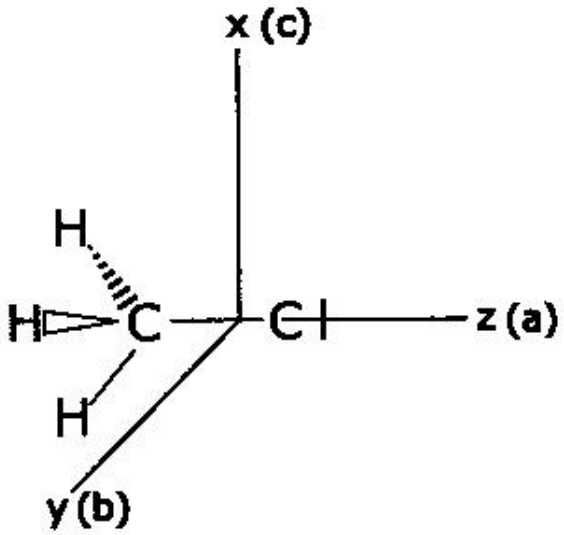
$I_C = I_B = I_A$

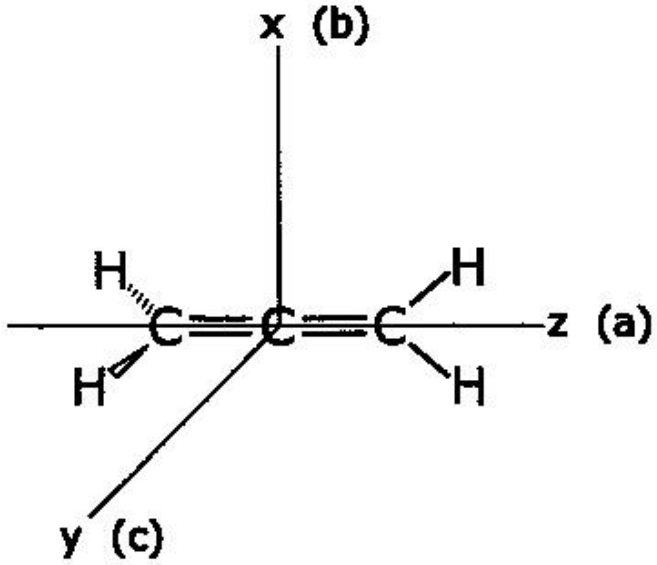
Spherical top

Other examples:

- CH_4
- CCl_4
- Generally, molecules with O_h, T_d, or I_h point groups are considered spherical top molecules.

Prolate Symmetric Top Molecules

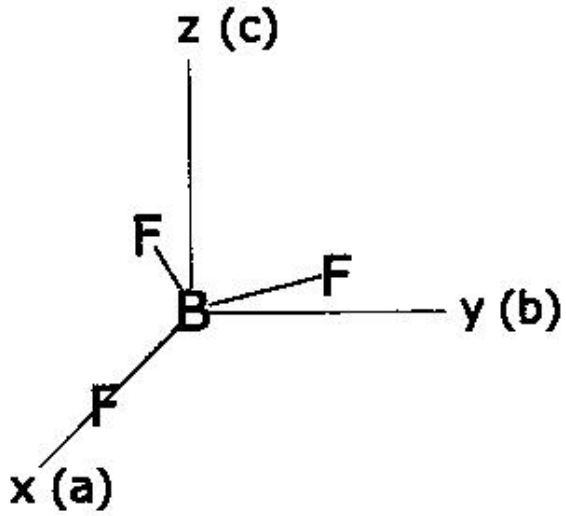


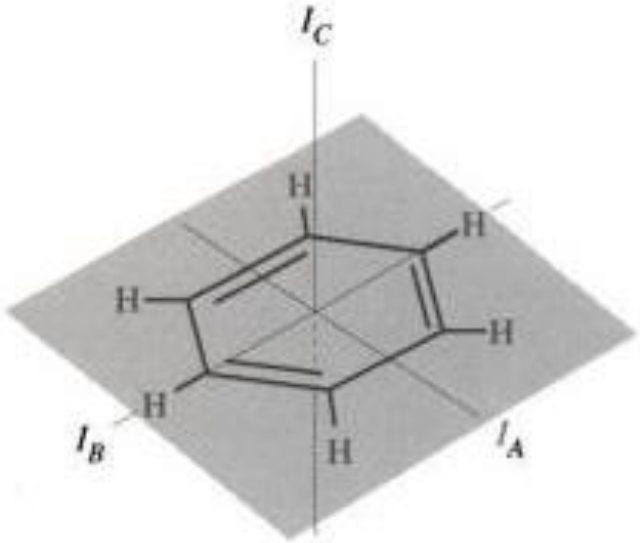


$$I_C = I_B > I_A$$

Prolate symmetric top

Oblate Symmetric Top Molecules

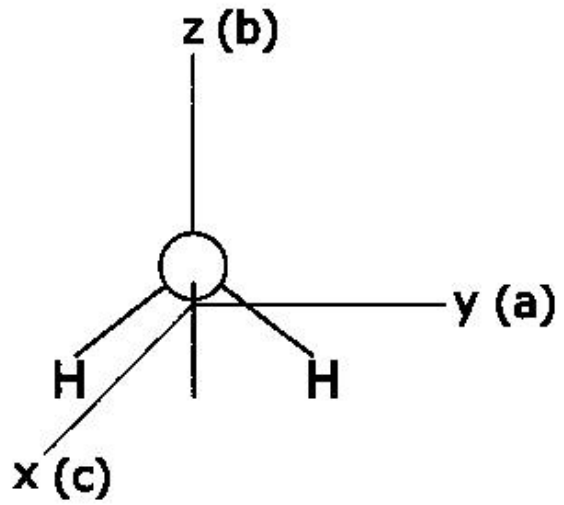




$$I_C = I_B < I_A$$

Oblate symmetric top

Asymmetric Top Molecules



Most of the molecules are asymmetric top.

$$I_C \neq I_B \neq I_A$$

Asymmetric top

- **Homonuclear diatomic molecules (such as H_2, O_2, N_2, Cl_2) – have zero dipole (non polar) - have zero change of dipole during the rotation, hence NO interaction with radiation - hence homonuclear diatomic molecules are microwave inactive.**

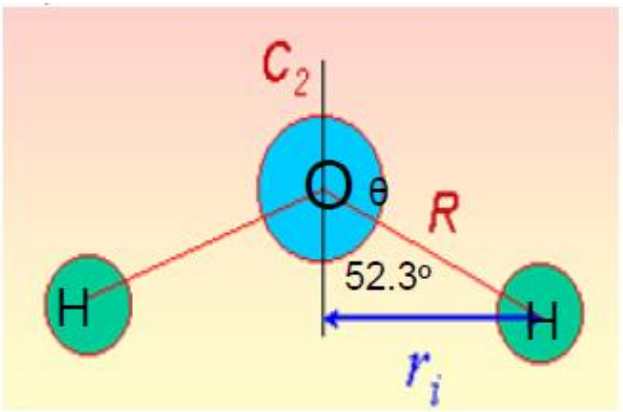
- **Heteronuclear diatomic molecules (such as HCl, HF, CO) – have permanent dipolemoment (polar compound) - change of dipole occurs during the rotation – hence interaction with radiation takes place – Therefore, heteronuclear diatomic molecules are microwave active.**

- **Calculate the moment of inertia of water molecule around the axis defined by the bisector of HOH bond. bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H = 1.0079, Atomic mass unit = 1.6606 x 10^{-27} kg).**

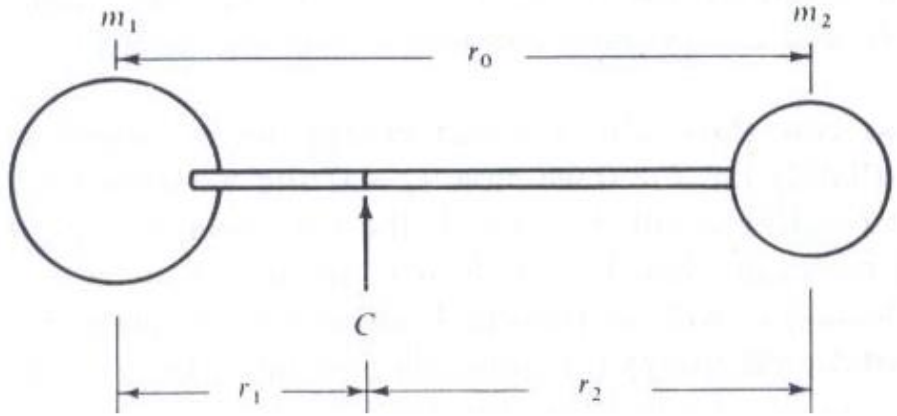
$$I = \sum_i m_i r_i^2 = m_H r_H^2 + 0 + m_H r_H^2 = 2\, m\, r^2 \sin^2\theta$$

$$= 2\times(1.67\times10^{-27}\ \text{kg}) \times (95.7\times10^{-12})^2 \times \text{Sin}^2\ 52.3$$

$$= 1.91\times 10^{-47}\ \text{kg m}^2$$



For linear diatomic molecules, the moment of inertia can be calculated as follows;



A rigid diatomic molecule treated as two masses, m_1 and m_2, joined by a rigid bar of length $r_0 = r_1 + r_2$

We start with this, the simplest of all linear molecules, shown in Fig. 2.1. Masses m_1 and m_2 are joined by a rigid bar (the bond) whose length is

$$r_0 = r_1 + r_2 \tag{2.5}$$

The molecule rotates end-over-end about a point C, the centre of gravity: this is defined by the moment, or balancing, equation:

$$m_1 r_1 = m_2 r_2 \tag{2.6}$$

The moment of inertia about C is defined by:

$$\begin{aligned} I &= m_1 r_1^2 + m_2 r_2^2 \\ &= m_2 r_2 r_1 + m_1 r_1 r_2 \qquad \text{(from (2.6))} \\ &= r_1 r_2 (m_1 + m_2) \end{aligned} \tag{2.7}$$

But, from (2.5) and (2.6):

$$m_1 r_1 = m_2 r_2 = m_2 (r_0 - r_1)$$

therefore,

$$r_1 = \frac{m_2 r_0}{m_1 + m_2} \qquad \text{and} \qquad r_2 = \frac{m_1 r_0}{m_1 + m_2} \tag{2.8}$$

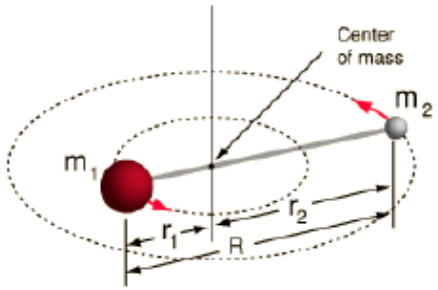
Replacing (2.8) into (2.7):

$$I = \frac{m_1 m_2}{m_1 + m_2} r_0^2 = \mu r_0^2 \tag{2.9}$$

where we have written $\mu = m_1 m_2/(m_1 + m_2)$, and μ is called the *reduced mass* of the system. Equation (2.9) defines the moment of inertia conveniently in terms of the atomic masses and the bond length.

Rotational Spectra of Linear Rigid Rotators

Simplest Case: Diatomic or Linear Polyatomic molecules
Rigid Rotor Model: Two nuclei are joined by a weightless rod



From solution of Schrodinger equation; $E_J = \frac{h^2}{8\pi^2 I} J(J+1)$

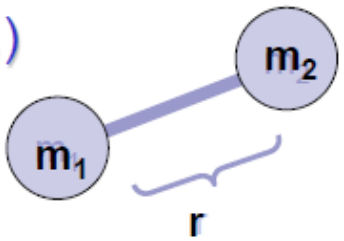
E_J = Rotational energy of rigid rotator (in Joules)

J = Rotational quantum number (J = 0, 1, 2, …)

I = Moment of inertia = μr^2

μ = reduced mass = $m_1 m_2 / (m_1 + m_2)$

r = internuclear distance (bond length)



Energy levels of a rigid diatomic rotor

1. Rotational energy for a rigid diatomic rotor E_J is quantized:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1)$$

units: *Joules*

J =0; 1; 2; . . . - rotational quantum number

2. Rotational energy is normally expressed in cm^{-1}

recall: $E = h \cdot \nu = h \cdot c \cdot \tilde{\nu}$ therefore: $\tilde{\nu} \equiv \varepsilon = \frac{E}{h \cdot c}$

$$\varepsilon_J = \frac{E_J}{hc} = \frac{h^2 J(J+1)}{4\pi^2 \cdot 2I \cdot hc} = \frac{h}{8\pi^2 I \cdot c} J(J+1)$$

B - rotational constant, units = cm^{-1}

$$\varepsilon_J = B \cdot J(J+1) \qquad B = \frac{h}{8\pi^2 I \cdot c}$$

Note: units for *c!*

B can be determined from spectroscopic experiments

Example

From microwave spectroscopy, bond lengths can be determined with a correspondingly high precision, as illustrated in this example. From the rotational microwave spectrum of $^{1}H^{35}Cl$, we find that B = 10.59342 cm^{-1}. Given that the masses of ^{1}H and ^{35}Cl are 1.0078250 and 34.9688527 amu, respectively, determine the bond length of the $^{1}H^{35}Cl$ molecule.

We have;

$$B = \frac{h}{8\pi^2 \mu c r_0^2}$$

$$r_0 = \sqrt{\frac{h}{8\pi^2 \mu c B}} = \sqrt{\frac{6.6260755 \times 10^{-34}}{8\pi^2 c \left(\frac{(1.0078250)(34.9688527)}{1.0078250 + 34.9688527}\right)(1.66054 \times 10^{-27})(10.59342)}}$$

$$= 1.274553 \times 10^{-10} m$$

Rotational transitions in rigid diatomic molecule.

Selection rules:

1- $\Delta \mu \neq 0$

molecule gives a rotational spectrum only if it has a permanent dipole moment

2- $\Delta J = \pm 1$

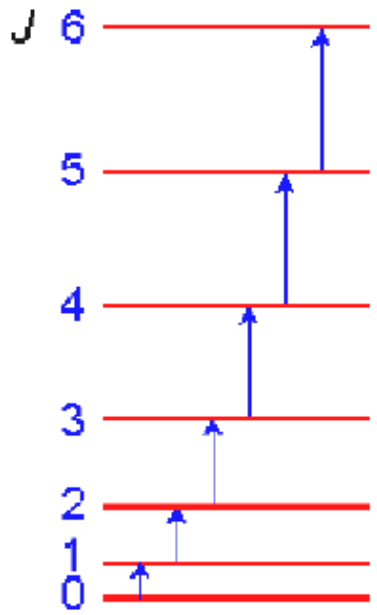
+1 absorption.

-1 emission.

Separation between adjacent levels:

$\Delta E_J = E(J) - E(J-1) = 2BJ$

and B can be obtained from the spacing between rotational lines in the spectra of molecules.



Allowed transitions

Transitions observed in the rotational spectrum

* For the transition; $J = 0 \rightarrow J = 1$ **Since;** $E_J = \dfrac{h}{8\pi^2 Ic} J(J+1)$

and; $B = h/8\,\pi^2\,Ic$

Then; $\Delta E_j = E_{j=1} - E_{j=0} = 2B - 0 = 2B\ cm^{-1}$

$= position\ \ of\ \ the\ \ first\ line\ in\ the\ spectrum = \overline{\upsilon}_{j=0 \rightarrow j=1}$

* For the transition; $J = 1 \rightarrow J = 2$

$\Delta E_j = E_{j=2} - E_{j=1} = 6B - 2B = 4B\ cm^{-1}$

$= position\ \ of\ \ the\ second\ line\ in\ the\ spectrum = \overline{\upsilon}_{j=1 \rightarrow j=2}$

* For the transition; $J = 2 \rightarrow J = 3$

$\Delta E_j = E_{j=3} - E_{j=2} = 12B - 6B = 6B\ cm^{-1}$

$= position\ \ of\ \ the\ third\ line\ in\ the\ spectrum = \overline{\upsilon}_{j=2 \rightarrow j=3}$

Since, The allowed rotational energies are given by;

$$\bar{\nu} = \frac{h}{8\pi^2 Ic} J(J+1)$$
$$= BJ(J+1)$$

The wave numbers of the different rotational levels will be; 0, 2B, 6B, 12B, 20B, 30B (cm^{-1}),… and so on

And for two adjacent rotational states, the energy difference is given by;

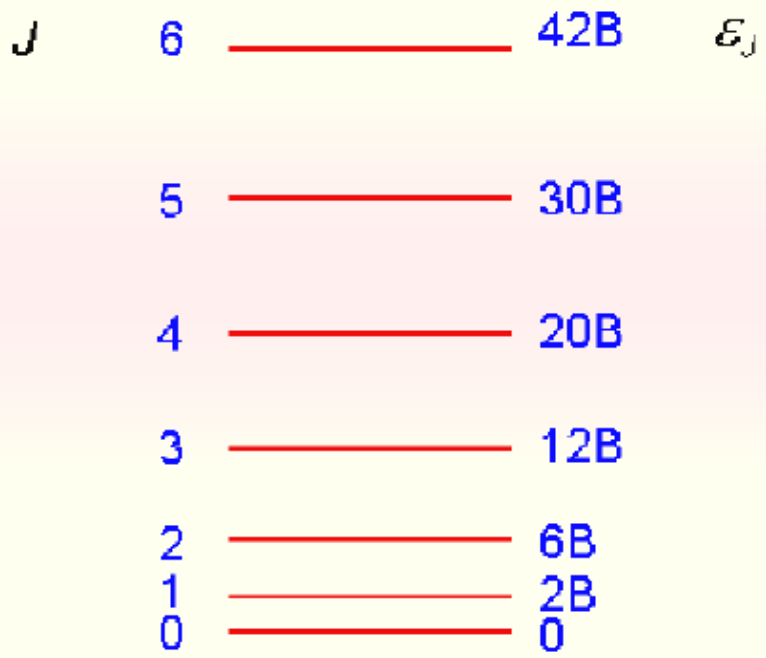
$$\Delta E_j = E_{j+1} - E_j = 2B(J+1)\ cm^{-1}$$

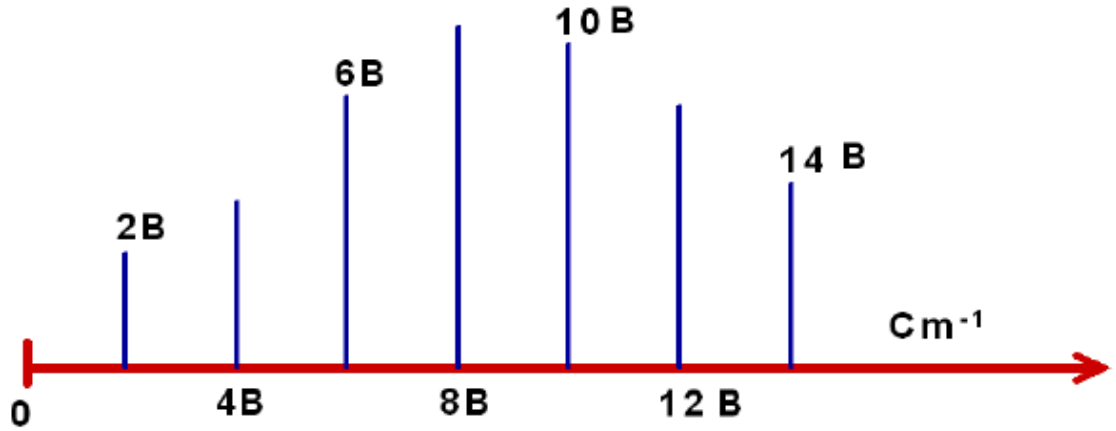
Hence, the wave number of the lines observed in the rotational spectrum will be; 2B, 4B, 6B, 8B (cm^{-1}), …… and so on.

And the various lines in the rotational spectra will be equally spaced (separation between lines = 2B)..

Energy levels of a rigid diatomic rotor

$$\varepsilon_J = B \cdot J(J+1) \qquad J=0;\ 1;\ 2;\ \ldots$$



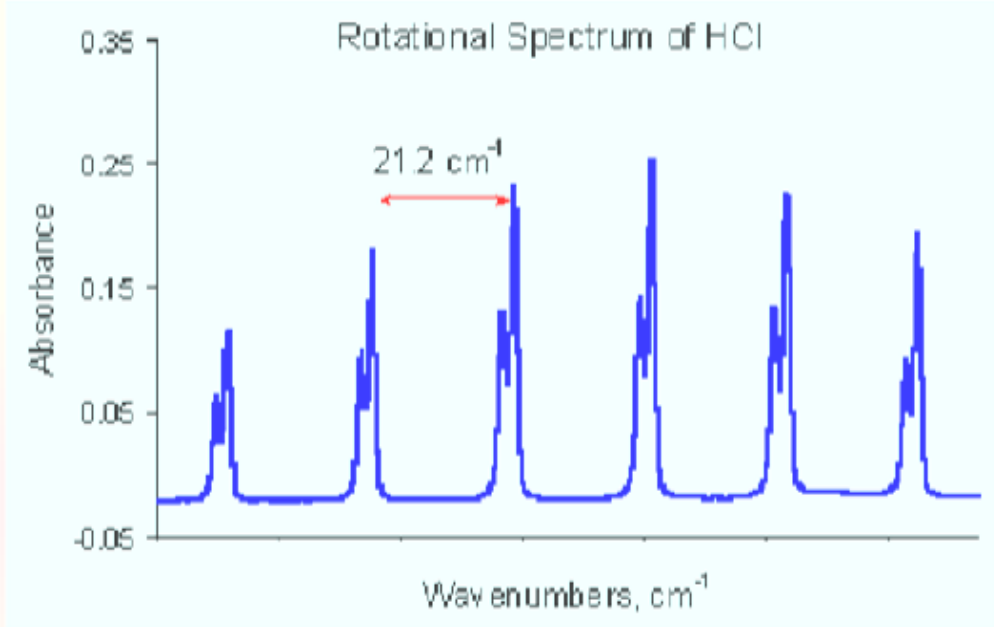


Microwave spectrum of rigid rotator

Separation between adjacent lines = 2B

So, B can be obtained from the spacing between rotational lines.

Examples of rotational spectra of rigid diatomic molecules



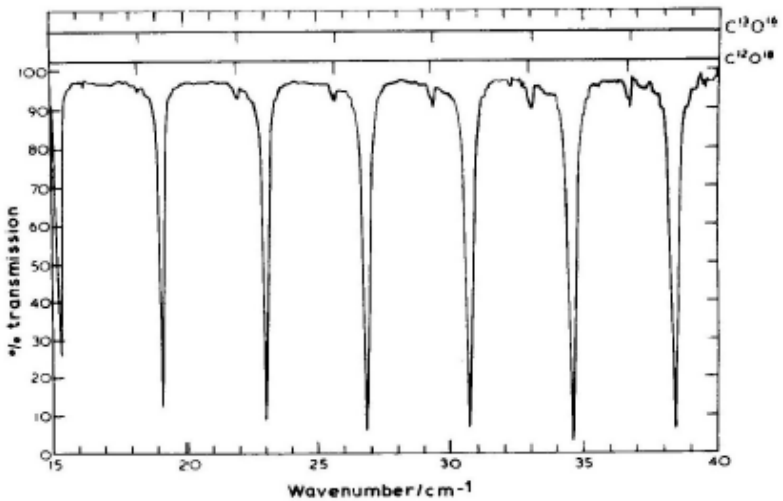
line separation in the rotational spectrum of HCl is $\approx 21.2\ cm^{-1} \Rightarrow B = 10.6\ cm^{-1}$; I_{HCl} can be found from:

$$B = \frac{h}{8\pi^2 I \cdot c}$$

next r_{HCl} can be obtained using
need to know m_x and m_y

$$I = \frac{m_1 \cdot m_2}{m_1 + m_2} \cdot r_0^2$$

Rotational Spectrum of CO



J" → J'	E(J')-E(J")	
3 → 4	2(1.91)(4)	15.3 cm^{-1}
4 → 5	2(1.91)(5)	19.1 cm^{-1}
5 → 6	2(1.91)(6)	22.9 cm^{-1}
6 → 7	2(1.91)(7)	26.7 cm^{-1}
7 → 8	2(1.91)(8)	30.6 cm^{-1}
8 → 9	2(1.91)(9)	34.4 cm^{-1}
9 → 10	2(1.91)(10)	38.2 cm^{-1}

J. Michael Hollas, *Modern Spectroscopy*, John Wiley & Sons, New York, 1992.

The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the rotational constant (B) and bond length of CO. The relative atomic weight C =12.00 and O = 15.9994, the absolute mass of H = 1.67343×10^{-27} kg.

$$\bar{\upsilon}_{0\to 1} = 3.8425 = 2\,B$$

$$B = 1.92118 \quad cm^{-1}$$

$$I = \frac{h}{8\pi^2\, B\, c}$$

$$I_{CO} = \frac{6.626 \times 10^{-34}}{8\pi^2 \times 2.99793 \times 10^{10} \times B}$$

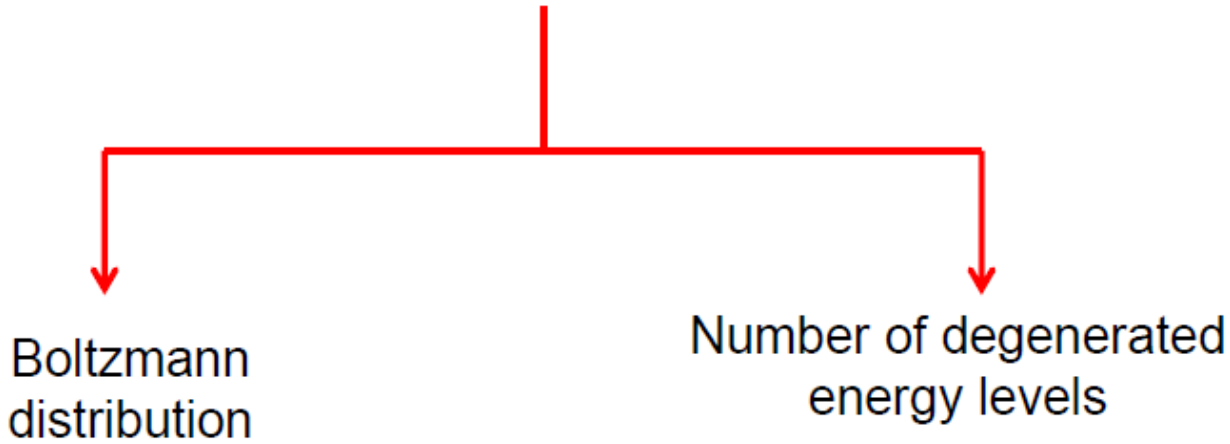
$$= 14.5695 \times 10^{-47} \quad kg\,.\,m^2$$

$$\mu = \frac{19.92168 \times 26.56136 \times 10^{-54}}{46.48303 \times 10^{-27}} = 11.38365 \quad kg$$

$$r^2 = \frac{I_{CO}}{\mu} = 1.2799 \times 10^{-20} \quad m^2$$

$$So, \qquad r_{CO} = 0.113 \quad nm \;=\; 1.131 \quad A^{o}$$

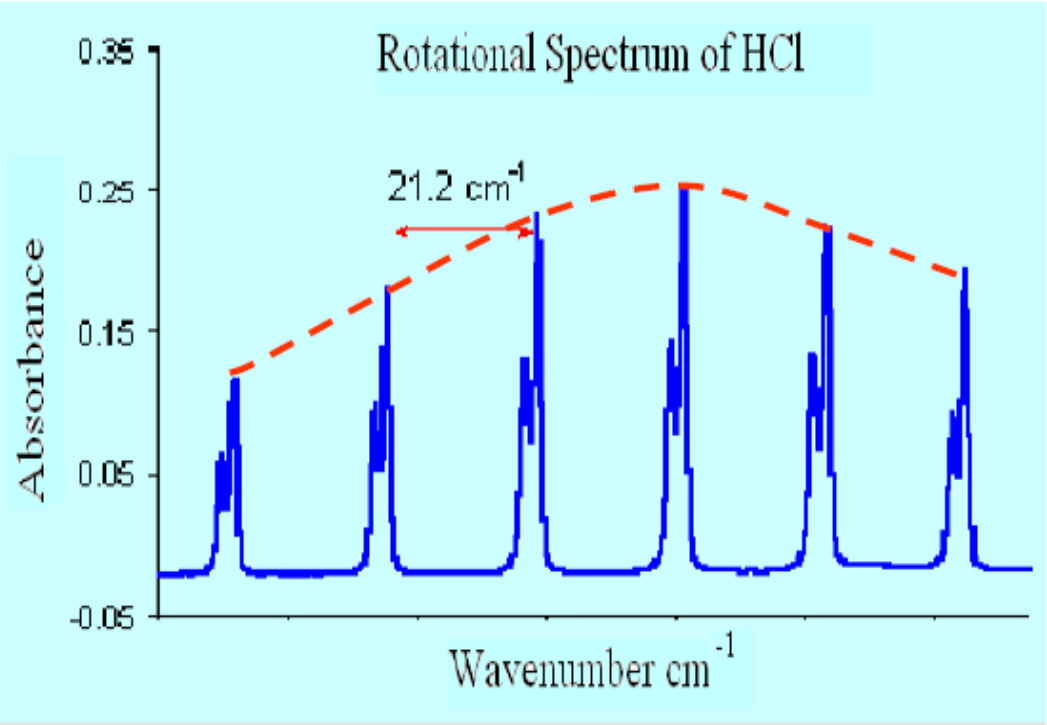
Intensity of rotational spectral lines (Population of energy levels)



Population of energy levels is affected by;

1- Boltzmann distribution
2- Number of degenerated energy levels (levels which have the same energies)

1-Boltzmann distribution



$$The\ number\ of\ molecules\ in\ the\ first\ state = N_o$$

$$The\ number\ of\ molecules\ in\ any\ higher\ state = N_j$$

$$\frac{N_j}{N_o} = e^{-E_J/kT} = e^{-(h\nu/kT)} = e^{-(hc\bar{\nu}/kT)}$$

$$\varepsilon_j = \bar{\nu} = BJ(J+1) \quad \text{So,}$$

$$\frac{N_j}{N_o} = e^{-[BhcJ(J+1)]/kT} = e^{-BJ(J+1)\,hc/kT}$$

The population of the state decreases as the J –value increases.

Example; For HCl, 2B = 21.2 → B = 10.6, T=300 K, h=6.626x10^{-34} J.s, k=1.38x10^{-23} J/K, c = 3x 10^{10} cm/s

$$J=0,\rightarrow \quad \frac{N_o}{N_o}=e^0=1$$

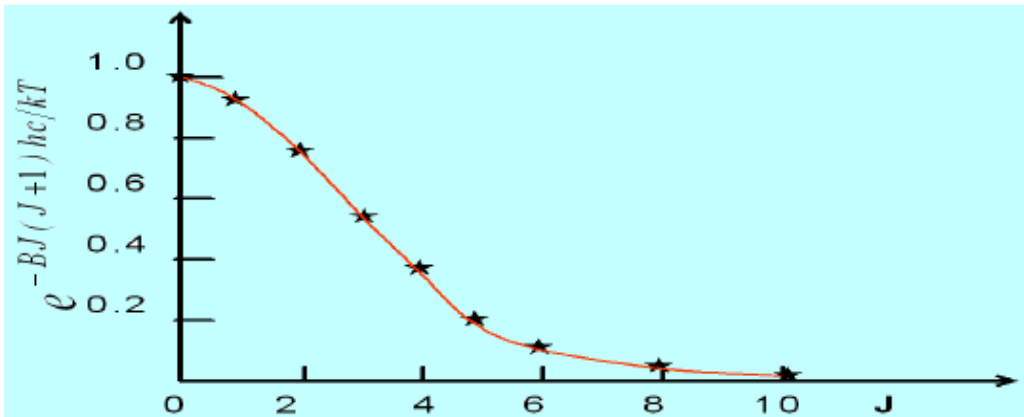
$$J=1,\rightarrow \quad \frac{N_1}{N_o}=e^{-\frac{10.6\times6.63\times10^{-34}\times3\times10^{10}}{1.38\times10^{-23}\times300}(J(J+1)}$$

$$=e^{-0.0509J(J+)}=0.903$$

$$J=2,\rightarrow \quad \frac{N_2}{N_o}=e^{-0.0509J(J+)}=0.7367$$

$$\frac{N_J}{N_o} \quad decreases \quad with \quad increa\sin g \; J$$

$$and \quad with \quad higher \quad B$$



The Boltzmann distribution alone does not fit the shape of the spectra so, degeneracy of the states is required.

2-Degeneracy

The existence of more than two energy states having the same energy.

Kinetic energy (K.E) in rotational motion = ½ $I.\omega^2$
and since, angular momentum (P) = $I.\omega$ where I is the moment of inertia, ω is the rotational frequency (in radians per second).

$$E = \frac{I \times I\omega^2}{2I} = \frac{I^2\,\omega^2}{2I} = \frac{P^2}{2I} \quad \Rightarrow \quad P = \sqrt{2\,E\,I}$$

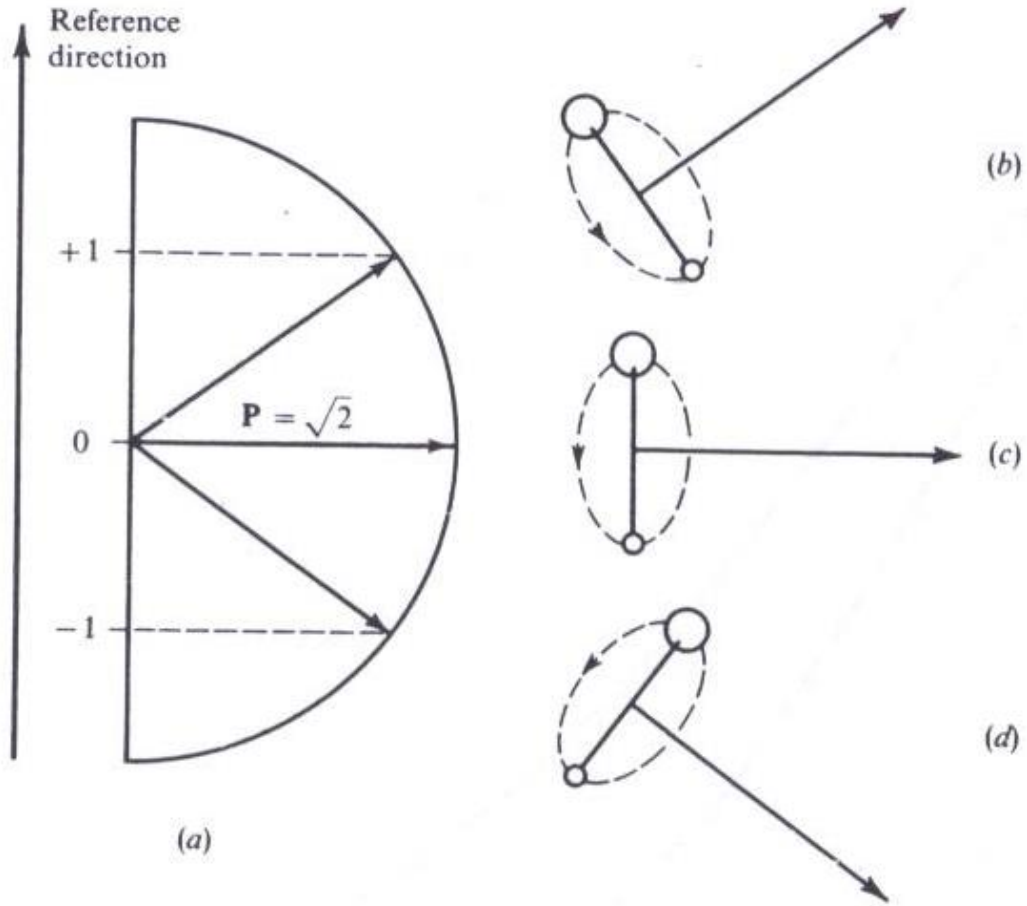
$$E = \frac{h^2}{4\pi^2} \frac{J(J+1)}{\times \quad 2I} \quad \Rightarrow \quad 2\,E\,I = \frac{h^2}{4\pi^2} \frac{J(J+1)}{} = \quad P^2$$

$$P = \sqrt{J\,(J+1)} \times \frac{h}{2\pi} = \sqrt{J\,(J+1)} \quad unit.$$

$\frac{h}{2\pi}$ *is the fundamental angular momentum unit.*

For J = 1

- **P is quantized, so it takes a directions such that the value of P along the reference direction are (-1, 0, +1).**
- **Number of degeneracy is 2J + 1 = 3**



We can see the implications of this most easily by means of a diagram. In Fig. 2.5 we show the case $J = 1$. Here $\mathbf{P} = \sqrt{1 \times 2}$ units $= \sqrt{2}$, and, as Fig. 2.5(a) shows, a vector of length $\sqrt{2}\,(= 1{\cdot}41)$ can have only *three* integral or zero components along a reference direction (here assumed to be from top to bottom in the plane of the paper): $+1$, 0, and -1. Thus the angular momentum vector in this instance can be oriented in only three different directions (Fig. 2.5(b)–(d)) with respect to the reference direction. All three rotational directions are, of course, associated with the same angular momentum and hence the same rotational energy: the $J = 1$ level is thus threefold degenerate.

Figure 2.5 The three degenerate orientations of the rotational angular momentum vector for a molecule with $J = 1$.

For J = 2

- **The value of P along the reference direction are (-2, -1, 0, +1, +2).**
- **Number of degeneracy is 2J + 1 = (2x2 + 1) = 5.**

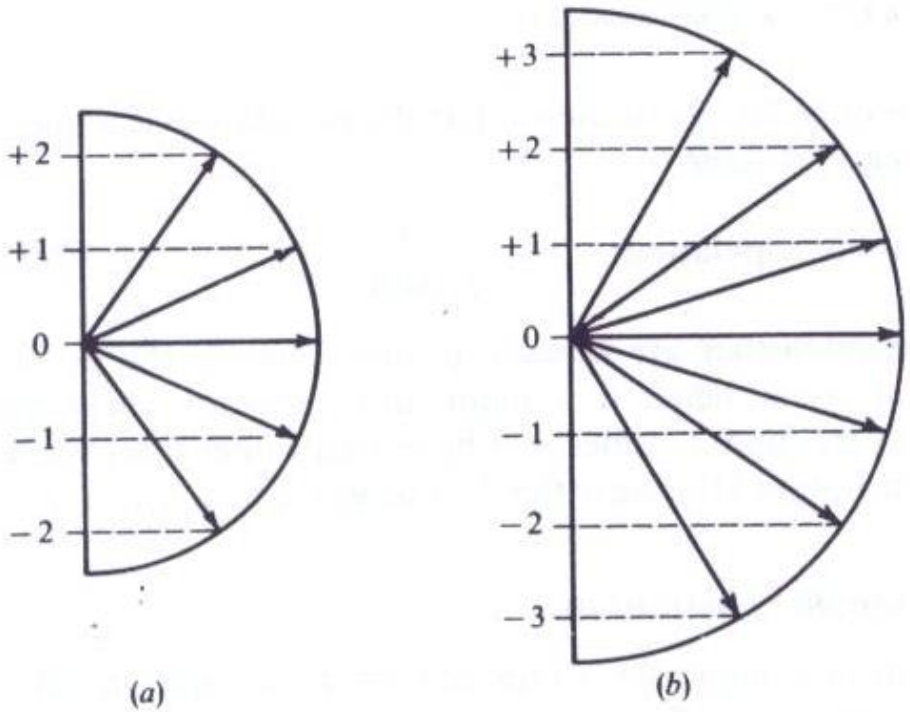


Figure 2.6 The five and seven degenerate rotational orientations for a molecule with $J = 2$ and $J = 3$ respectively.

Figure 2.6(*a*) and (*b*) shows the situation for $J = 2$ ($\mathbf{P} = \sqrt{6}$) and $J = 3$ ($\mathbf{P} = 2\sqrt{3}$) with fivefold and sevenfold degeneracy respectively. In general it may readily be seen that *each energy level is* $2J + 1$*-fold degenerate.*

Thus we see that, although the molecular population in each level decreases exponentially (Eq. (2.18)), the number of degenerate levels available increases rapidly with J. The total relative population at an energy E_J will plainly be:

$$\text{Population} \propto (2J + 1) \exp(-E_J/kT) \tag{2.20}$$

When this is plotted against J the points fall on a curve of the type shown in Fig. 2.7, indicating that the population rises to a maximum and then

$$N_J/N_0 = \exp(-E_J/kT) = \exp\{-BhcJ(J + 1)/kT\} \tag{2.18}$$

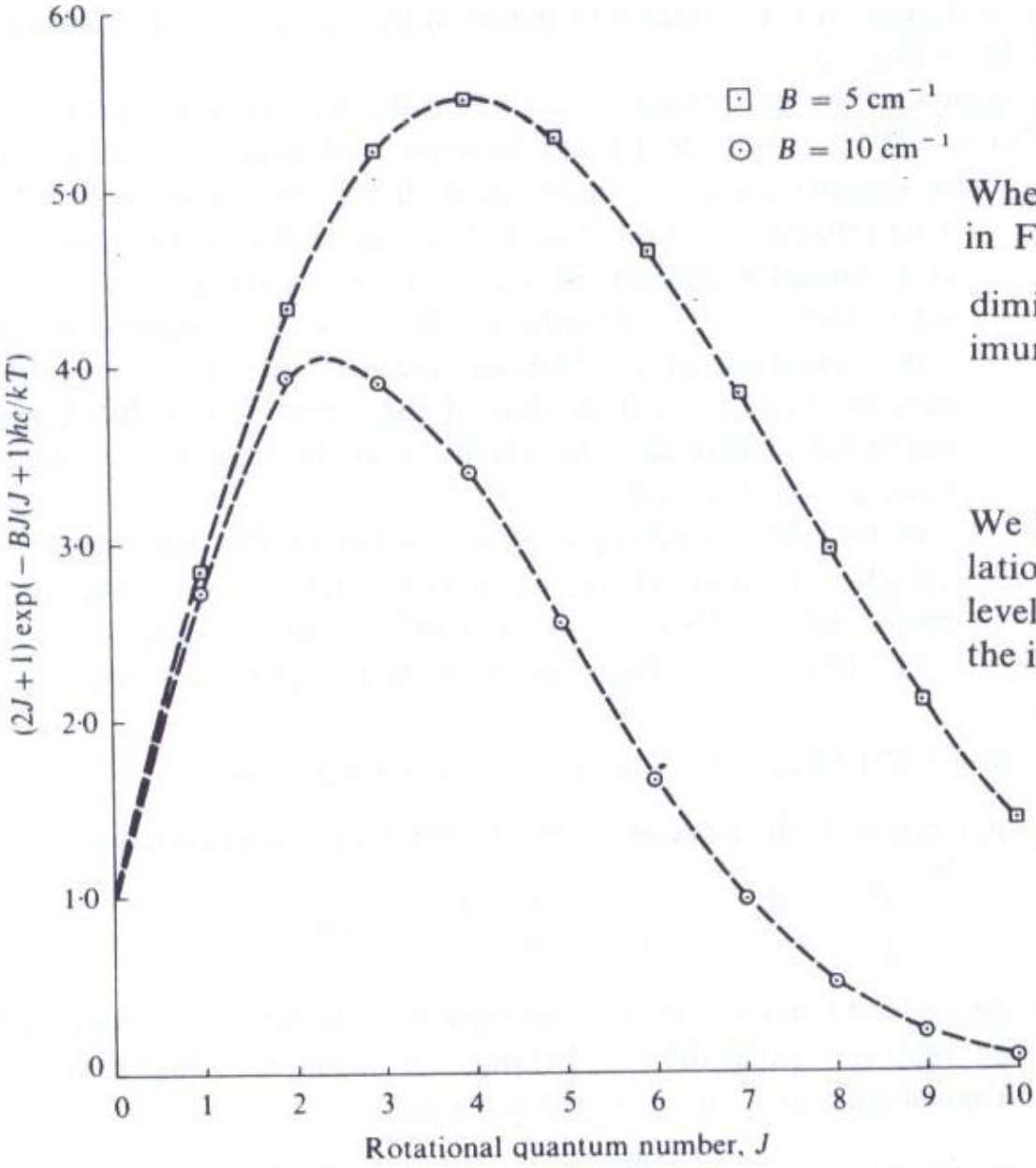


Figure 2.7 The total relative populations, including degeneracy, of the rotational energy levels of a diatomic molecule. The diagram has been drawn for the same conditions as Fig. 2.4.

When this is plotted against J the points fall on a curve of the type shown in Fig. 2.7, indicating that the population rises to a maximum and then diminishes. Differentiation of Eq. (2.20) shows that the population is a maximum at the nearest integral J value to:

$$\text{Maximum population:} \quad J = \sqrt{\frac{kT}{2hcB}} - \frac{1}{2} \tag{2.21}$$

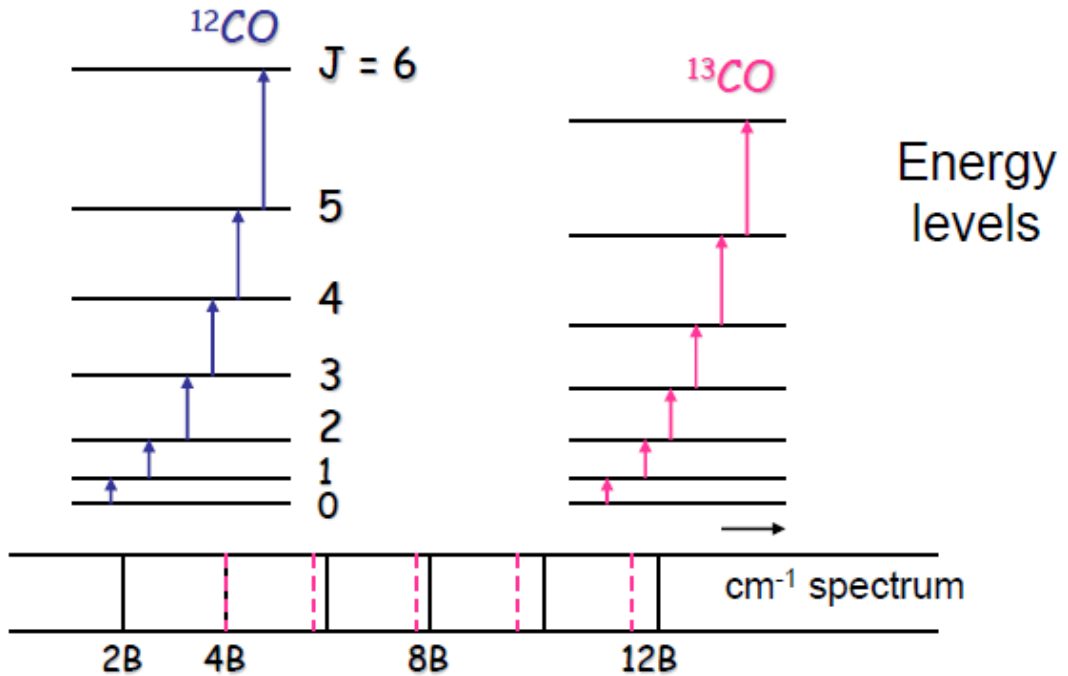
We have seen that line intensities are directly proportional to the populations of the rotational levels, hence it is plain that transitions between levels with very low or very high J values will have small intensities while the intensity will be a maximum at or near the J value given by Eq. (2.21).

Bond Length and Rotational Constants of Diatomic Molecules

	r_o, Å	B, cm^{-1}		r_o, Å	B, cm^{-1}
H_2^+	1.06	29.8	F_2	1.42	0.88
H_2	0.74138	60.8	Cl_2	1.98	0.24
D_2	0.74154	30.4	Br_2	2.28	0.081
O_2	1.21	1.44	HF	0.92	20.9
N_2	1.10	2.00	HCl	1.27	10.6
CO	1.13	1.93	HBr	1.41	8.46

Effect of isotopic substitution

On changing from $^{12}C^{16}O \rightarrow {}^{13}C^{16}O$, atomic mass increases, B decreases ($\propto$ 1/I), so energy of levels becomes lower.



From comparison of rotational energy levels of ^{12}CO and ^{13}CO

We can determine:

(i) isotopic masses accurately, to within 0.02% of other methods for atoms in gaseous molecules;

(ii) isotopic abundances from the absorption relative intensities.

Example:

for ^{12}CO the transition J=0 → J=1 appears at 3.84235 cm^{-1}

for ^{13}CO the transition J=0 → J=1 appears at 3.67337 cm^{-1}

Given : ^{12}C = 12.0000 ; O = 15.9994 amu

What is isotopic mass of ^{13}C ?

$B(^{12}CO) = 1.921175\ cm^{-1}$

$B(^{13}CO) = 1.836685\ cm^{-1}$

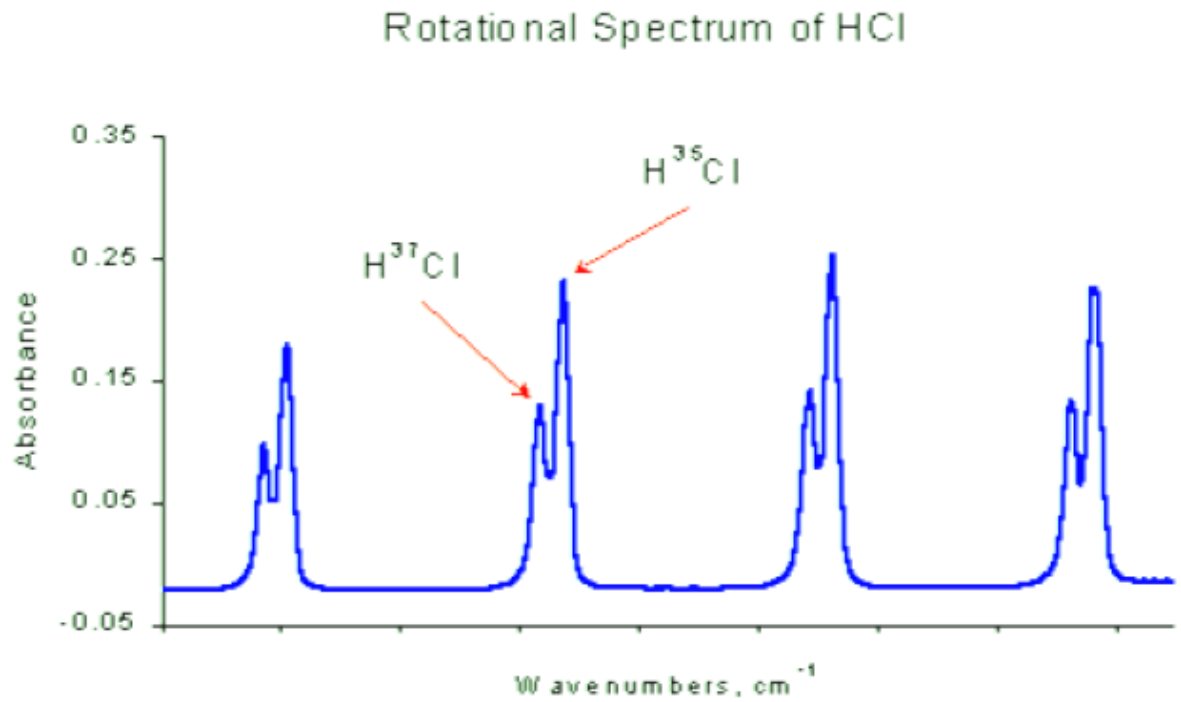
Now $$B \propto \frac{1}{I} \propto \frac{1}{\mu}$$

$$\Rightarrow \frac{\mu(^{13}CO)}{\mu(^{12}CO)} = \frac{1.921175}{1.836685} = 1.04600$$

$$\Rightarrow 1.046 = \frac{(^{13}C)\times 15.9994}{(^{13}C)+15.9994} \times \frac{12+15.9994}{12\times 15.9994}$$

$$\Rightarrow (^{13}C) = 13.0006 \text{ amu}$$

Effect of Isotopic Substitution



We can obtain; atomic masses, bond length or relative abundance of isotopes

2.3.5 The Spectrum of a Non-Rigid Rotator

The Schrödinger wave equation may be set up for a non-rigid molecule, and the rotational energy levels are found to be:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) - \frac{h^4}{32\pi^4 I^2 r^2 k} J^2(J+1)^2 \text{ J}$$

or

$$\varepsilon_J = E_J/hc = BJ(J+1) - DJ^2(J+1)^2 \text{ cm}^{-1} \tag{2.24}$$

where the rotational constant, B, is as defined previously, and the *centrifugal distortion constant* D, is given by:

$$D = \frac{h^3}{32\pi^4 I^2 r^2 kc} \text{ cm}^{-1} \tag{2.25}$$

which is a positive quantity. Equation (2.24) applies for a simple harmonic force field only; if the force field is anharmonic, the expression becomes:

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + KJ^4(J+1)^4 \cdots \text{ cm}^{-1} \tag{2.26}$$

where H, K, etc., are small constants dependent upon the geometry of the molecule. They are, however, negligible compared with D and most modern spectroscopic data are adequately fitted by Eq. (2.24).

From the defining equations of B and D it may be shown directly that

$$D = \frac{16B^3\pi^2\mu c^2}{k} = \frac{4B^3}{\bar{\omega}^2} \tag{2.27}$$

where $\bar{\omega}$ is the vibrational frequency of the bond, and k has been expressed according to Eq. (2.22).

Non-rigid Rotor

For a non-rigid rotor, the bond length increases as the angular velocity increases

$$B = \frac{h}{8\pi^2 I \cdot c} \Rightarrow B \propto \frac{1}{r^2}$$

$\Rightarrow$ *B* decreases with *J*

Rotational transitions of HF

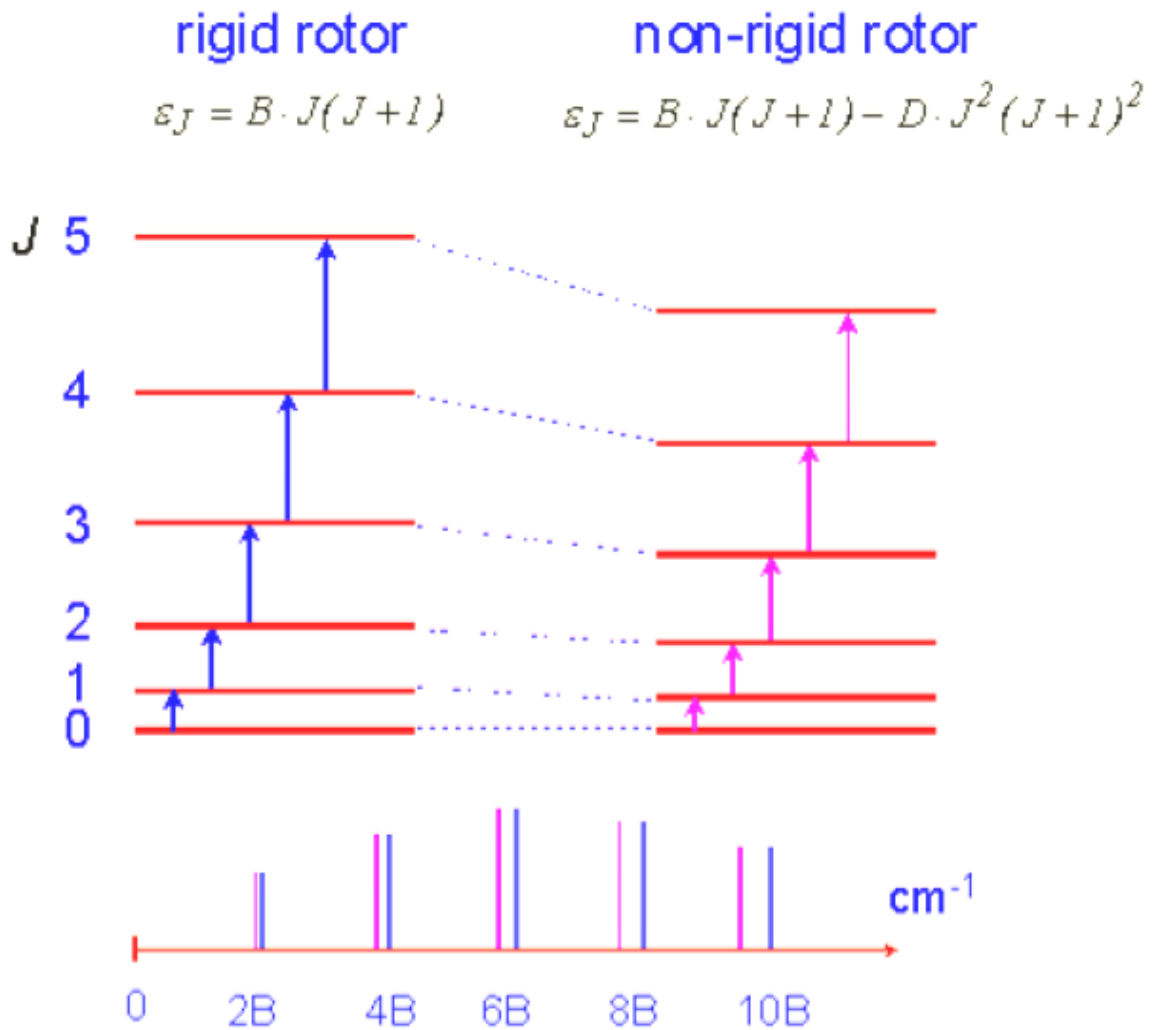
Transition	B_{calc} cm^{-1}	B_{exp} cm^{-1}	r, Å
0→1	20.56	20.55	0.929
1→2	20.56	20.48	0.931
2→3	20.56	20.43	0.932
3→4	20.56	20.31	0.935
4→5	20.56	20.16	0.938
5→6	20.56	20.04	0.941
6→7	20.56	19.82	0.946
7→8	20.56	19.64	0.951
8→9	20.56	19.45	0.955
9→10	20.56	19.16	0.963
10→11	20.56	18.91	0.969

for a non-rigid rotor

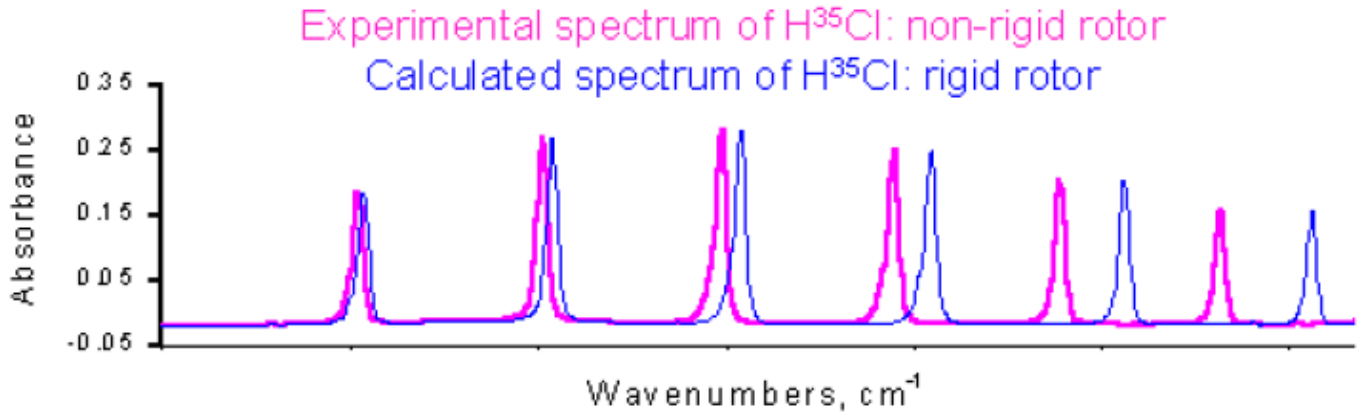
$$\varepsilon_J = B \cdot J(J+1) - D \cdot J^2(J+1)^2$$

D – centrifugal distortion constant ('stiffness' constant)

units: cm^{-1}


rigid rotor
non-rigid rotor
$\varepsilon_J = B \cdot J(J+1)$
$\varepsilon_J = B \cdot J(J+1) - D \cdot J^2 (J+1)^2$
J
5
4
3
2
1
0
cm-1
0
2B
4B
6B
8B
10B

Non-rigid Rotor



$$\varepsilon_J = B \cdot J(J+1) - D \cdot J^2 (J+1)^2$$

$$\tilde{\nu} = \varepsilon_{J+1} - \varepsilon_J = 2B(J+1) - 4D \cdot (J+1)^3$$

the rigid rotor selection rules apply, *i.e.* $\Delta J = \pm 1$

$B \approx 1$ to 10 cm^{-1} $\quad$ $D \approx 10^{-3}$ to 10^{-2} cm^{-1}

2.4 POLYATOMIC MOLECULES

2.4.1 Linear Molecules

We consider first molecules such as carbon oxysulphide OCS, or chloroacetylene HC≡CCl, where all the atoms lie on a straight line, since this type gives rise to a particularly simple spectra in the microwave region. Since $I_B = I_C$; $I_A = 0$, as for diatomic molecules, the energy levels are given by a formula identical with Eq. (2.26), i.e.,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + \cdots \text{ cm}^{-1} \tag{2.30}$$

and the spectrum will show the same $2B$ separation modified by the distortion constant. In fact, the whole of the discussion on diatomic molecules applies equally to all linear molecules; three points, however, should be underlined:

1. Since the moment of inertia for the end-over-end rotation of a polyatomic linear molecule is considerably greater than that of a diatomic molecule, the B value will be much smaller, and the spectral lines more closely spaced. Thus B values for diatomic molecules are about 10 cm^{-1}, while for triatomic molecules they can be 1 cm^{-1} or less, and for larger molecules smaller still.
2. The molecule must, as usual, possess a dipole moment if it is to exhibit a rotational spectrum. Thus OCS will be microwave active, while OCO (more usually written CO_2) will not. In particular, it should be noted that isotopic substitution does not lead to a dipole moment since the bond lengths and atomic charges are unaltered by the substitution. Thus $^{16}OC^{18}O$ is microwave inactive.

3. A non-cyclic polyatomic molecule containing N atoms has altogether $N - 1$ individual bond lengths to be determined. Thus in the triatomic molecule OCS there is the CO distance, r_{CO}, and the CS distance, r_{CS}. On the other hand, there is only *one* moment of inertia for the end-over-end rotation of OCS, and only this one value can be determined from the spectrum. Table 2.2 shows the data for this molecule. Over the four lines observed there is seen to be no appreciable centrifugal distortion, and, taking the value of B as 0·2027 cm^{-1}, we calculate:

$$I_B = \frac{h}{8\pi^2 Bc} = 137{\cdot}95 \times 10^{-47} \text{ kg m}^2$$

From this one observation it is plainly impossible to deduce the two unknowns, r_{CO} and r_{CS}. The difficulty can be overcome, however, if we study a molecule with different atomic masses but the *same* bond lengths—i.e., an isotopically substituted molecule—since this will have a different moment of inertia.

Table 2.2 Microwave spectrum of carbon oxysulphide

$J \rightarrow J+1$	$\bar{\nu}_{obs.}$ (cm^{-1})	$\Delta\bar{\nu}$	B (cm^{-1})
0 → 1	...		
		2 × 0·4055	0·2027
1 → 2	0·8109		
		0·4054	0·2027
2 → 3	1·2163		
		0·4054	0·2027
3 → 4	1·6217		
		0·4054	0·2027
4 → 5	2·0271		

Let us consider the rotation of OCS in some detail. Figure 2.10 shows the molecule, where r_O, r_C, and r_S represent the distances of the atoms from the centre of gravity. Consideration of moments gives:

$$m_O r_O + m_C r_C = m_S r_S \tag{2.31}$$

where m_i is the mass of atom i. The moment of inertia is:

$$I = m_O r_O^2 + m_C r_C^2 + m_S r_S^2 \tag{2.32}$$

and we have the further equations:

$$r_O = r_{CO} + r_C \qquad r_S = r_{CS} - r_C \tag{2.33}$$

where r_{CO} and r_{CS} are the bond lengths of the molecule. It is these we wish to determine. Substituting (2.33) in (2.31) and collecting terms:

$$(m_C + m_O + m_S)r_C = m_S r_{CS} - m_O r_{CO}$$

or

$$Mr_C = m_S r_{CS} - m_O r_{CO} \tag{2.34}$$

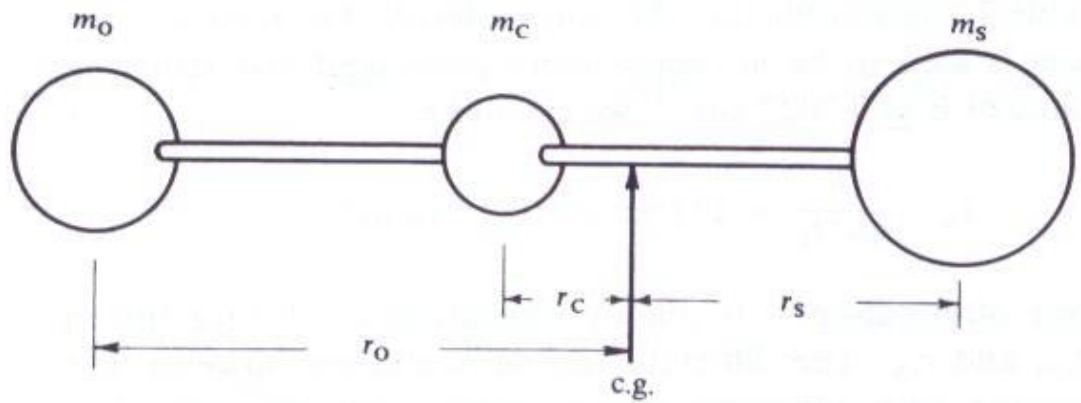


Figure 2.10 The molecule of carbon oxysulphide, OCS, showing the distances of each atom from the centre of gravity.

where we write M for the total mass of the molecule. Substituting (2.33) in (2.32):

$$\begin{aligned} I &= m_O(r_{CO} + r_C)^2 + m_C r_C^2 + m_S(r_{CS} - r_C)^2 \\ &= Mr_C^2 + 2r_C(m_O r_{CO} - m_S r_{CS}) + m_O r_{CO}^2 + m_S r_{CS}^2 \end{aligned}$$

and finally substituting for r_C from Eq. (2.34):

$$I = m_O r_{CO}^2 + m_S r_{CS}^2 - \frac{(m_O r_{CO} - m_S r_{CS})^2}{M} \tag{2.35}$$

Considering now the isotopic molecule, ^{18}OCS, we may write m'_O for m_O throughout Eq. (2.35):

$$I' = m'_O r_{CO}^2 + m_S r_{CS}^2 - \frac{(m'_O r_{CO} - m_S r_{CS})^2}{M'} \tag{2.36}$$

and we can now solve for r_{CO} and r_{CS}, provided we have extracted a value for I' from the microwave spectrum of the isotopic molecule. Note that we do *not* need to write r'_{CO}, since we assume that the bond length is unaltered by isotopic substitution. This assumption may be checked by studying the molecules $^{16}OC^{34}S$ and $^{18}OC^{34}S$, since we would then have four moments of inertia. The bond distances found are quite consistent, and hence justify the assumption.

The extension of the above discussion to molecules with more than three atoms is straightforward; it suffices to say here that microwave studies have led to very precise determinations of many bond lengths in such molecules.

2.4.2 Symmetric Top Molecule

Although the rotational energy levels of this type of molecule are more complicated than those of linear molecules, we shall see that, because of their symmetry, their pure rotational spectra are still relatively simple. Choosing methyl fluoride again as our example we remember that

$$I_B = I_C \neq I_A \qquad I_A \neq 0$$

There are now two directions of rotation in which the molecule might absorb or emit energy—that about the main symmetry axis (the C—F bond in this case) and that perpendicular to this axis.

We thus need two quantum numbers to describe the degree of rotation, one for I_A and one for I_B or I_C. However, it turns out to be very convenient mathematically to have a quantum number to represent the *total* angular momentum of the molecule, which is the sum of the separate angular momenta about the two different axes. This is usually chosen to be the quantum number J. Reverting for a moment to *linear* molecules, remember that we there used J to represent the end-over-end rotation of a molecule: however, this was the *only* sort of rotation allowed, so it is quite consistent to use J, in general, to represent the *total angular momentum.* It is then conventional to use K to represent the angular momentum about the top axis—i.e., about the C—F bond in this case.

Let us briefly consider what values are allowed to K and J. Both must, by the conditions of quantum mechanics, be integral or zero. The total angular momentum can be as large as we like, that is, J can be 0, 1, $2, \ldots, \infty$ (except, of course, for the theoretical possibility that a real molecule will be disrupted at very high rotational speeds). Once we have chosen J, however, K is rather more limited. Let us consider the case when $J = 3$. Plainly the rotational energy can be divided in several ways between motion about the main symmetry axis and motion perpendicular to this. If *all* the rotation is about the axis, $K = 3$; but note that K cannot be greater than J since J is the *total* angular momentum. Equally we could have $K = 2$, 1, or 0, in which case the motion perpendicular to the axis increases accordingly. Additionally, however, K can be negative—we can imagine positive and negative values of K to correspond with clockwise and anti-clockwise rotation about the symmetry axis—and so can have values -1, -2, or -3.

In general, then, for a total angular momentum J, we see that K can take values:

$$K = J, J-1, J-2, \ldots, 0, \ldots, -(J-1), -J \tag{2.37}$$

which is a total of $2J + 1$ values altogether. This figure of $2J + 1$ is important and will recur.

If we take first the case of a *rigid* symmetric top—i.e., one in which the bonds are supposed not to stretch under centrifugal forces—the Schrödinger equation may be solved to give the allowed energy levels for rotation as:

$$\varepsilon_{J,K} = E_{J,K}/hc = BJ(J+1) + (A-B)K^2 \quad \text{cm}^{-1} \tag{2.38}$$

where, as before,

$$B = \frac{h}{8\pi^2 I_B c} \qquad \text{and} \qquad A = \frac{h}{8\pi^2 I_A c}$$

Note that the energy depends on K^2, so that it is immaterial whether the top spins clockwise or anticlockwise: the energy is the same for a given angular momentum. For all $K > 0$, therefore, the rotational energy levels are *doubly degenerate.*

The selection rules for this molecule may be shown to be:

$$\Delta J = \pm 1 \text{ (as before)} \qquad \text{and} \qquad \Delta K = 0 \tag{2.39}$$

and, when these are applied to Eq. (2.38), the spectrum is given by:

$$\begin{aligned} \varepsilon_{J+1,K} - \varepsilon_{J,K} = \bar{\nu}_{J,K} &= B(J+1)(J+2) + (A-B)K^2 \\ &\quad -[BJ(J+1) + (A-B)K^2] \\ &= 2B(J+1) \text{ cm}^{-1} \end{aligned} \tag{2.40}$$

Thus the spectrum is independent of K, and hence rotational changes about the symmetry axis do not give rise to a rotational spectrum. The reason for this is quite evident—rotation about the symmetry axis does not change the dipole moment perpendicular to the axis (which always remains zero), and hence the rotation cannot interact with radiation. Equation (2.40) shows that the spectrum is just the same as for a linear molecule and that only one moment of inertia—that for end-over-end rotation—can be measured.

Both Eqs (2.38) and (2.40) are for a rigid molecule, however, and we have already seen that microwave spectroscopy is well able to detect the departure of real molecules from this idealized state. When centrifugal stretching is taken into account, the energy levels become:

$$\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2 - D_J J^2(J+1)^2 - D_{JK} J(J+1)K^2 - D_K K^4 \quad \text{cm}^{-1} \quad (2.41)$$

where, in an obvious notation, D_J, D_{JK}, and D_K are small correction terms for non-rigidity. The selection rules are unchanged (Eq. (2.39)), and so the spectrum is:

$$\begin{aligned} \bar{\nu}_{J,K} &= \varepsilon_{J+1,K} - \varepsilon_{J,K} \\ &= 2B(J+1) - 4D_J(J+1)^3 - 2D_{JK}(J+1)K^2 \quad \text{cm}^{-1} \end{aligned} \quad (2.42)$$

We see that the spectrum will be basically that of a linear molecule (including centrifugal stretching) with an additional term depending on K^2.

It is easy to see why this spectrum now depends on the axial rotation (i.e., depends on K), although such rotation produces no dipole change. Figure 2.11 illustrates methyl fluoride for (*a*) $K = 0$, no axial rotation, and (*b*) $K > 0$, the molecule rotating about the symmetry axis. We see, from the much exaggerated diagram, that axial rotation widens the HCH angles and stretches the C—H bonds. The distorted molecule (*b*) has a different moment of inertia for end-over-end rotation from (*a*). If we write Eq. (2.42) as:

$$\bar{\nu}_{JK} = 2(J+1)[B - 2D_J(J+1)^2 - D_{JK}K^2] \quad \text{cm}^{-1}$$

we can see more clearly that the centrifugal distortion constants D_J and D_{JK} can be considered as correction terms to the rotational constant B, and hence as perturbing the moment of inertia I_B.

Since each value of J is associated with $2J + 1$ values of K, we see that each line characterized by a certain J value must have $2J + 1$ components.

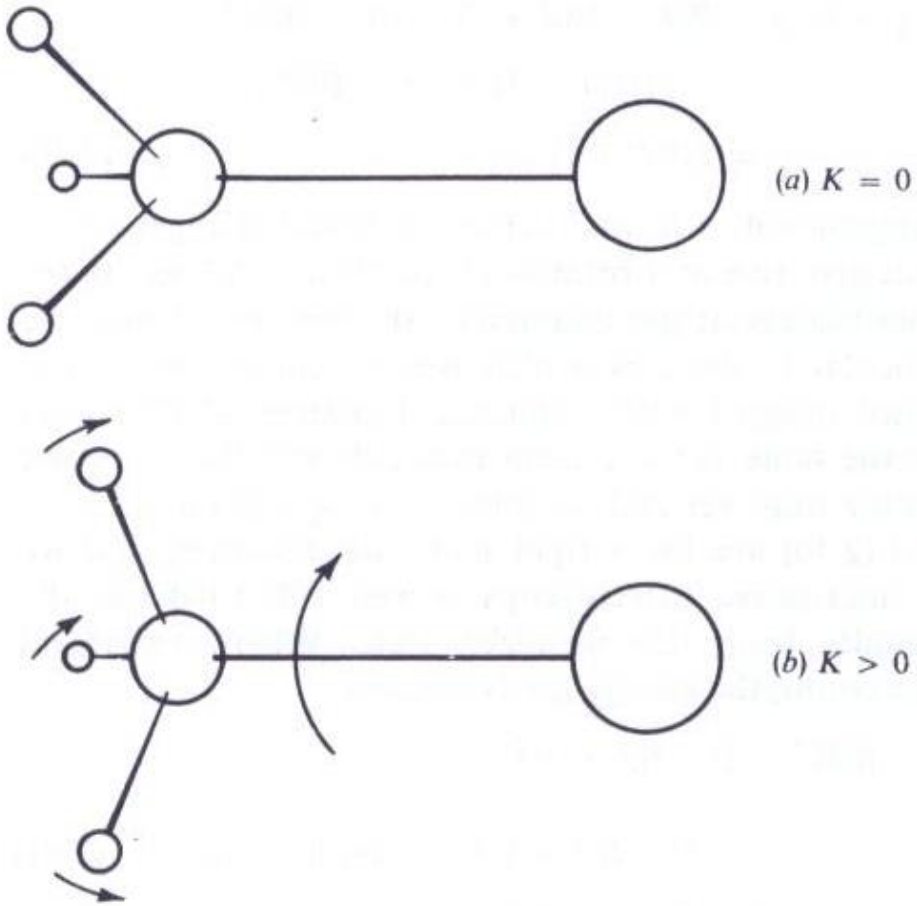


Figure 2.11 The influence of axial rotation on the moment of inertia of a symmetric top molecule, e.g., methyl fluoride, CH_3F. In (*a*) there is no axial rotation ($K = 0$), and in (*b*) $K > 0$.

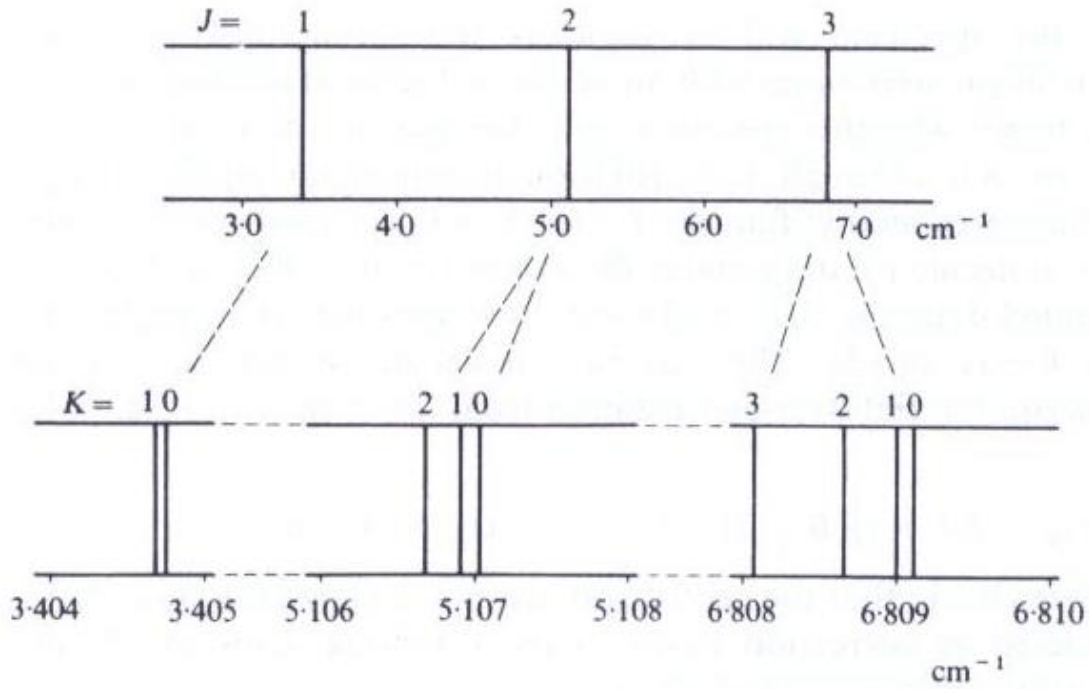


Figure 2.12 A diagrammatic representation of the rotational spectrum of the symmetric top molecule methyl fluoride, CH_3F.

However, since K only appears as K^2 in Eq. (2.42), there will be only $J + 1$ *different* frequencies, all those with $K > 0$ being doubly degenerate. We may tabulate a few lines as follows:

$$\begin{aligned}
J = 0,\ K = 0 \qquad & \bar{\nu}_{JK} = 2B - 4D_J \quad \text{cm}^{-1} \\
J = 1,\ K = 0 \qquad & \bar{\nu}_{JK} = 4B - 32D_J \\
K = \pm 1 \qquad & \bar{\nu}_{JK} = 4B - 32D_J - 4D_{JK} \\
J = 2,\ K = 0 \qquad & \bar{\nu}_{JK} = 6B - 108D_J \\
K = \pm 1 \qquad & \bar{\nu}_{JK} = 6B - 108D_J - 6D_{JK} \\
K = \pm 2 \qquad & \bar{\nu}_{JK} = 6B - 108D_J - 24D_{JK}, \quad \text{etc.}
\end{aligned} \tag{2.43}$$

Let us now compare these with the observed spectrum of methyl fluoride. This is shown as a line diagram in Fig. 2.12, and the frequencies are tabulated in Table 2.3. Fitting these data to equations such as (2.43) leads directly to:

$$B = 0{\cdot}851\,204 \text{ cm}^{-1}$$
$$D_J = 2{\cdot}00 \times 10^{-6} \text{ cm}^{-1}$$
$$D_{JK} = 1{\cdot}47 \times 10^{-5} \text{ cm}^{-1}$$

The calculated frequencies of Table 2.3 show how precisely such measurements may now be made.

Once again each spectrum examined yields only one value of B, but the spectra of isotopic molecules can, in principle, give sufficient information for the calculation of all the bond lengths and angles of symmetric top molecules, together with estimates of the force constant of each bond.

Table 2.3 Microwave spectrum of methyl fluoride

J	K	$\bar{\nu}_{\text{obs.}}$ †(cm^{-1})	$\bar{\nu}_{\text{calc.}}$ (cm^{-1})
1	0	3·404 75	3·404 752
	1	3·404 70	3·404 693
2	0	5·107 01	5·107 008
	1	5·106 92	5·106 920
	2	5·106 65	5·106 655
3	0	6·809 12	6·809 120
	1	6·809 00	6·809 002
	2	6·808 65	6·808 649
	3	6·808 06	6·808 062

† Taken from W. Gordy, *Physical Review*, **93**, 406 (1954), by kind permission of the author.

Table 2.4 Some molecular data determined by microwave spectroscopy

Molecule	Type	Bond length (nm)	Bond angle (deg.)	Dipole moment† (debyes)
NaCl	Diatomic	0·236 06 ± 0·000 01	—	8·5 ± 0·2
COS	Linear	0·116 4 ± 0·000 1 (CO) 0·155 9 ± 0·000 1 (CS)	—	0·712 ± 0·004
HCN	Linear	0·106 317 ± 0·000 005 (CH) 0·115 535 ± 0·000 006 (CN)	—	2·986 ± 0·004
NH_3	Sym. Top	0·100 8 ± 0·000 4	107·3 ± 0·2	1·47 ± 0·01
CH_3Cl	Sym. Top	0·109 59 ± 0·000 05 (CH) 0·178 12 ± 0·000 05 (CCl)	108·0 ± 0·2 (HCH)	1·871 ± 0·005
H_2O	Asym. Top	0·095 84 ± 0·000 05	104·5 ± 0·3	1·846 ± 0·005
O_3	Asym. Top	0·127 8 ± 0·000 2	116·8 ± 0·5	0·53 ± 0·02

† Measured from the Stark effect, cf. Sec. 2.5.2.

Microwave spectroscopy follows the usual pattern: source, monochromator, beam direction, sample and detector.

1. *The source and monochromator.* The usual source in this region is the klystron valve which, since it emits radiation of only a very narrow frequency range, is called 'monochromatic' and acts as its own monochromator. The actual emission frequency is variable electronically and hence a spectrum may be scanned over a limited range of frequencies using a single klystron.

   One slight disadvantage of this source is that the total energy radiated is very small—of the order of milliwatts only. However, since all this is concentrated into a narrow frequency band a sharply tuned detector can be sufficiently activated to produce a strong signal.
2. *Beam direction.* This is achieved by the use of 'waveguides'—hollow tubes of copper or silver, usually of rectangular cross-section—inside which the radiation is confined. The waveguides may be gently tapered or bent to allow focusing and directing of the radiation. Atmospheric absorption of the beam is considerable, so the system must be efficiently evacuated.
3. *Sample and sample space.* In almost all microwave studies so far the sample has been gaseous. However, pressures of 0·01 mmHg are sufficient to give a reasonable absorption spectrum, so many substances which are usually thought of as solid or liquid may be examined provided their vapour pressures are above this value. The sample is retained by very thin mica windows in a piece of evacuated waveguide.
4. *Detector.* It is possible to use an ordinary superheterodyne radio receiver as detector, provided this may be tuned to the appropriate high frequency; however, a simple crystal detector is found to be more sensitive and easier to use. This detects the radiation focused upon it by the waveguide, and the signal it gives is amplified electronically for display on an oscilloscope, or for permanent record on paper.

The Stark Effect

Experimentally the Stark effect requires the placing of an electric field, either perpendicular or parallel to the direction of the radiation beam, across the sample. Practically it is simpler to have a perpendicular field. We shall consider three advantages of this field.

1. A molecule exhibiting a rotational spectrum must have an electric dipole moment, and so its rotational energy levels will be perturbed by the application of an exterior field since interaction will occur. Put simply, the absorption lines of the spectrum will be shifted by an amount depending on the extent of the interaction, and thus depending on both E, the applied field, and μ, the dipole moment. For a linear molecule the shift is found to be:

$$\Delta\nu \propto (\mu E)^2 \qquad \text{(linear molecule)}$$

while for a symmetric top:

$$\Delta\nu \propto \mu E \qquad \text{(symmetric top)}$$

Thus we have immediately a very accurate method of determining dipole moments, simply by observation of the Stark shift. More important, the measurement is made on very dilute gas samples, so the dipole moment observed may be taken to be that of the actual molecule, uncomplicated by molecular interactions, solvent effects, etc. Some values determined in this way are included in Table 2.4.

2. The second valuable application of the Stark effect is in the assignment of observed spectral lines to particular J values. We have seen that, in the absence of marked departure from rigidity and good resolving power, the assignment of J values is not always obvious. The line of lowest frequency which we observe *may* happen to correspond with $J = 0$, or it may be that it is the first observable line of a series, either because earlier lines are intrinsically very weak or because of limitations in the apparatus used. However, we have seen that each line is $2J + 1$ degenerate because rotations can occur in $2J + 1$ orientations in space without violating quantum laws. In the absence of any orienting effect these transitions have precisely the same frequency, but a Stark field constitutes an orienting effect, and splits the degeneracy; thus multiplet structure is observed for all lines with $J > 0$. The number of components depends on J, and hence unambiguous assignments can be made.
3. The final application is purely an instrumental one, but is especially interesting in that it has its counterpart in other spectral regions. We have already referred to the concept of signal-to-noise ratio in Chapter 1; that part of the noise which arises from random fluctuations in the background radiation may be removed by modulating the beam by means of the Stark effect as explained below.

Imagine the application of a Stark field in a periodic manner such as the 'square-wave' variation of Fig. 2.13; while the field is switched on the signal is modified in the way described in 1 and 2 above. If we arrange the modulation frequency to be some 100 to 1000 Hz, and construct the amplifier so that it amplifies only the component of the signal which has the modulated frequency, stray radiation which has not been through the modulating field will be completely ignored. This results in a great improvement of the signal-to-noise ratio.

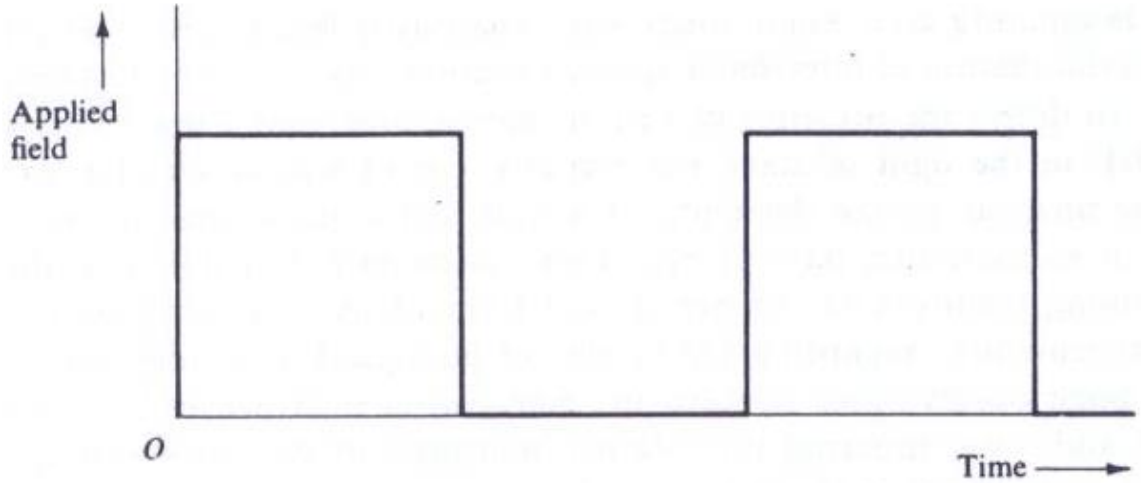


Figure 2.13 A 'square wave' potential as used for Stark modulation.